

Discrepancy functions and using them in calibration

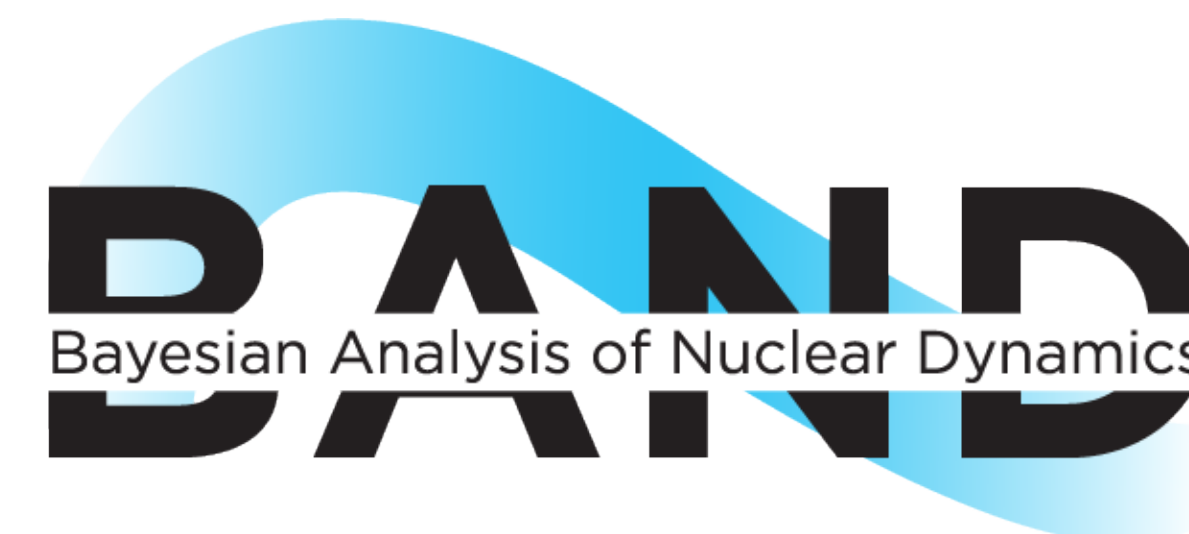
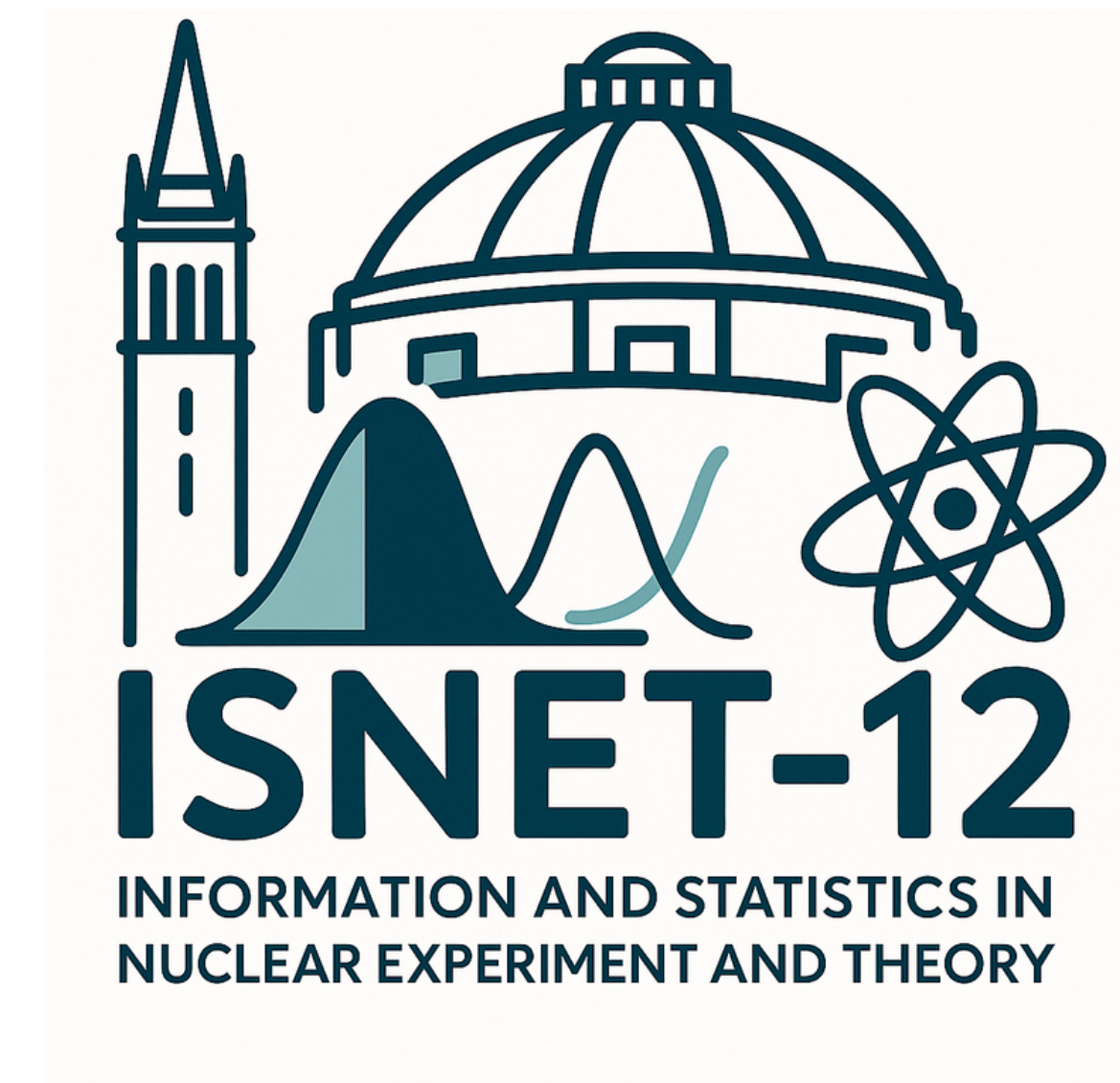
Sunil Jaiswal and Dick Furnstahl

ISNET-12: BAND Camp

Link to tutorial materials: <https://go.osu.edu/bandcamp>

Based on BAND Example: ModelDiscrepancy

October 5, 2026



Setup for BAND Camp tutorial

To do in your browsers:

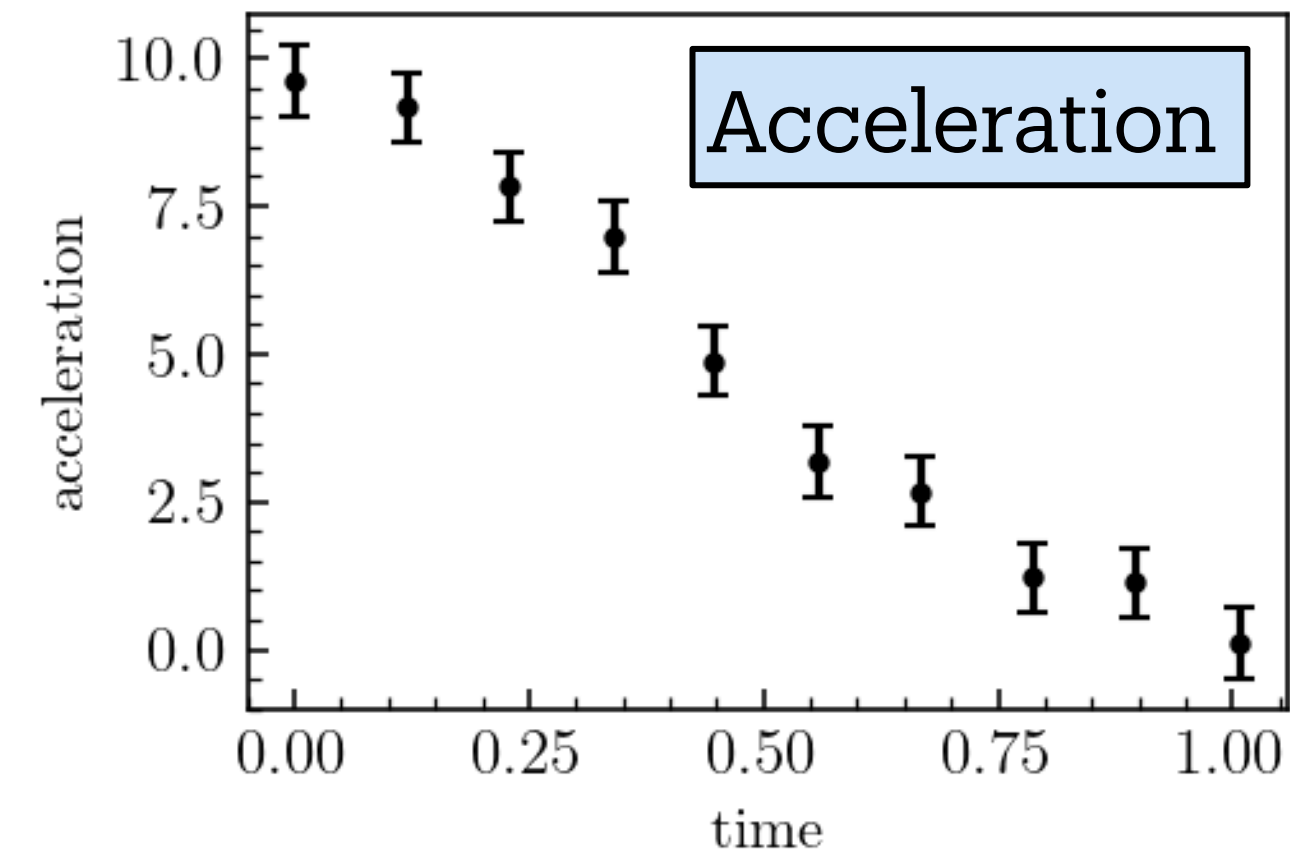
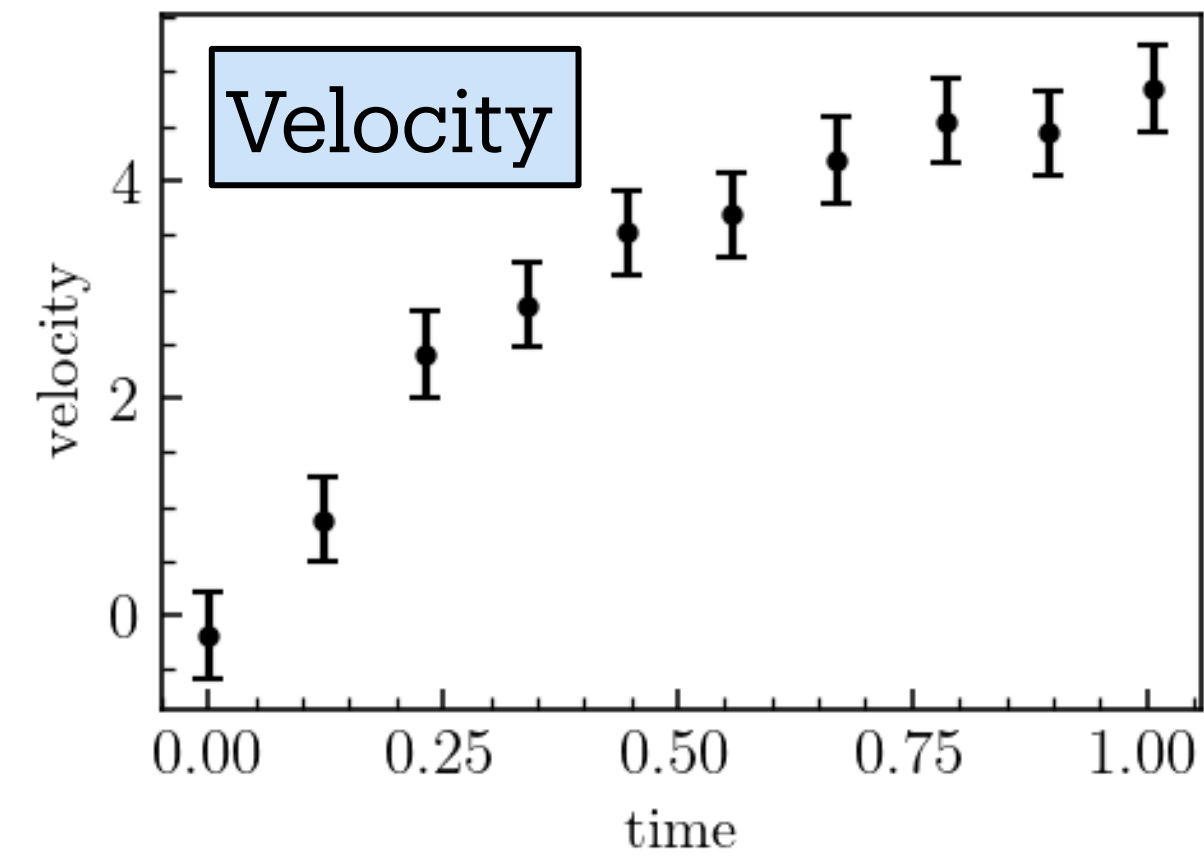
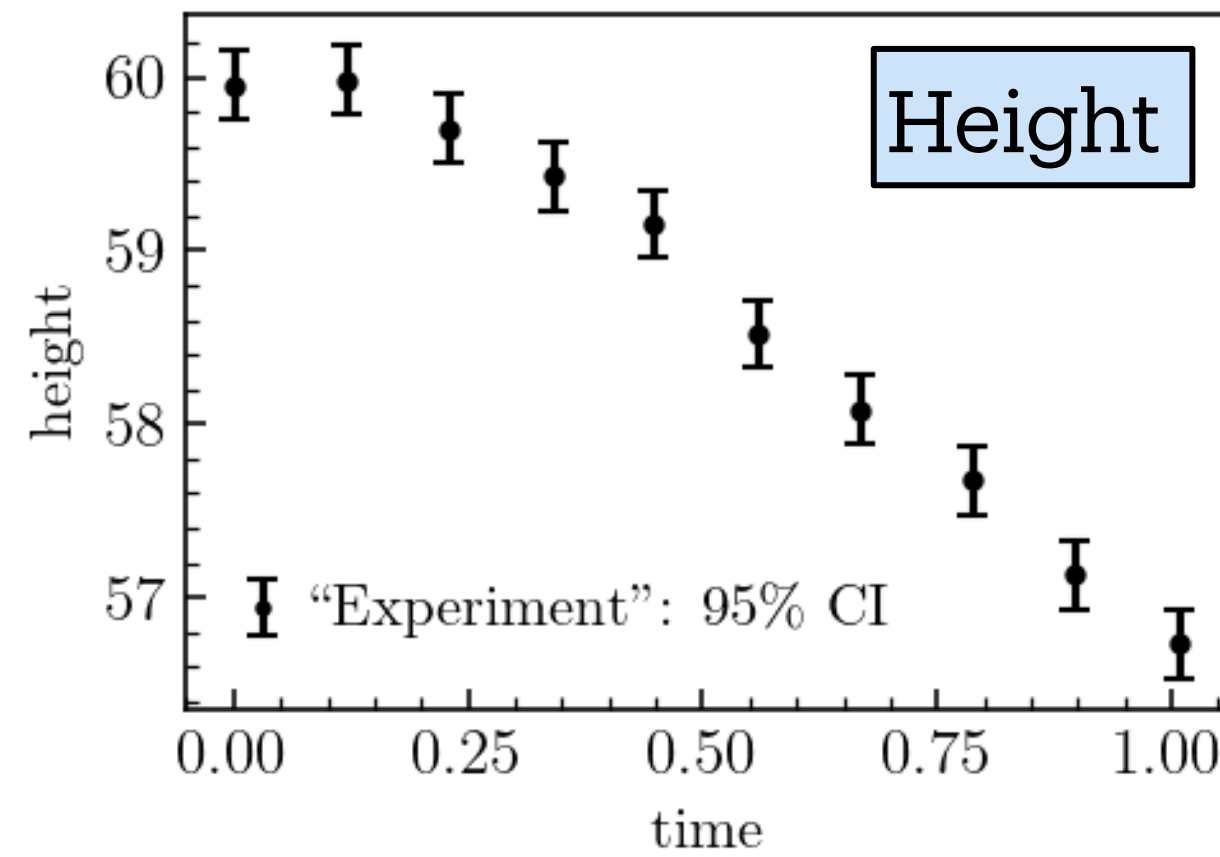
- Go to the README at <https://go.osu.edu/bandcamp> and open the *Demo: Ball-drop experiment* notebook in Google Colab (there is a link; open in a new tab)
- **Run only the pip install cell and the three cells in A1:** *Generate "experimental" observations for height.* **Tell the facilitator if you encounter any problems.**

From the BAND repository:

- We won't be generating results from the BAND repository in this session (takes too long to run), but you can run notebooks in Colab that make plots using pre-calculated results.
- From the README: Ball-drop experiment results from [arXiv:2504.13144](https://arxiv.org/abs/2504.13144)
- From the README: Heavy-ion discrepancy model results from [arXiv:2509.19759](https://arxiv.org/abs/2509.19759)

Motivation: A simple experiment

A ball is dropped from a tower. Height, velocity, acceleration are measured with time.



TRUTH

Drag force: $\mathbf{f}_D = -0.4v^2 \hat{\mathbf{v}}$

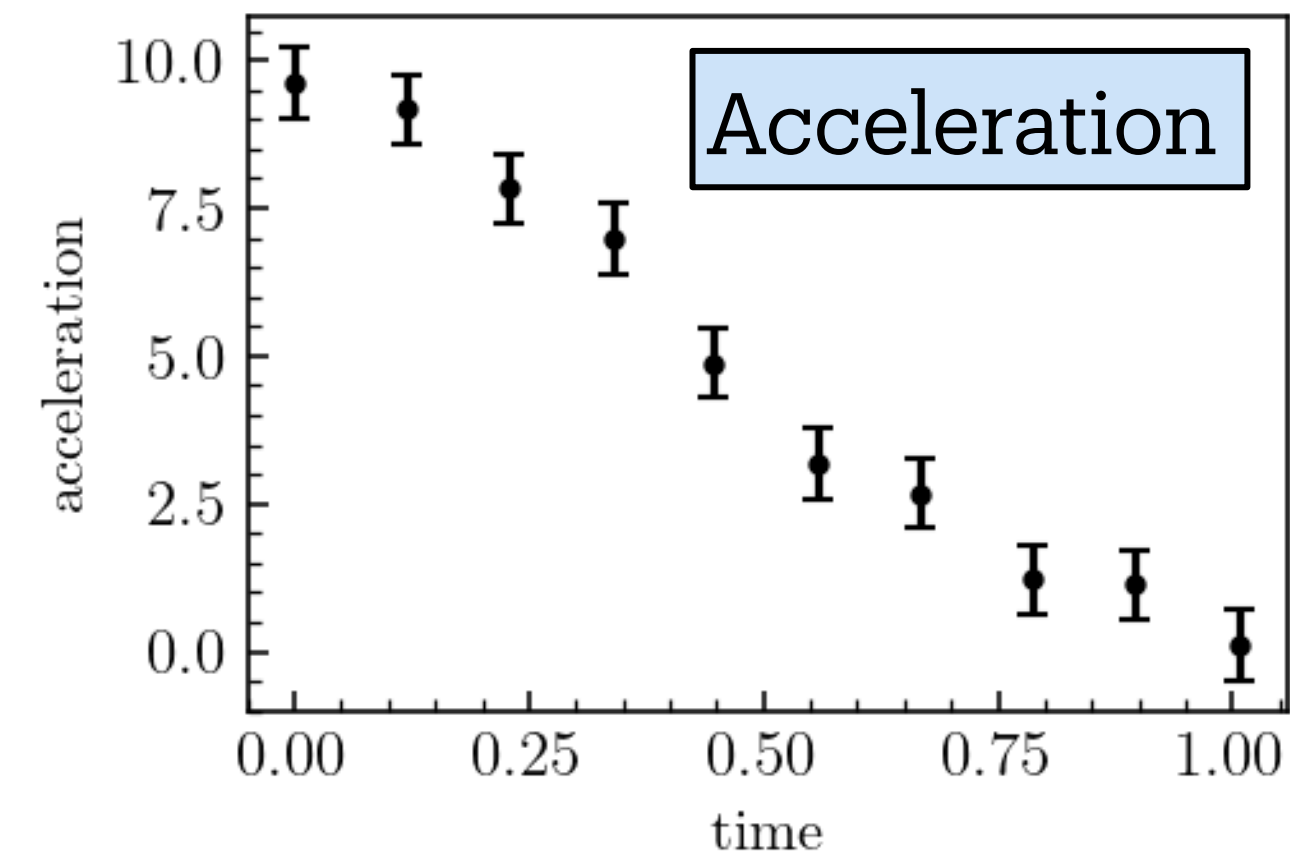
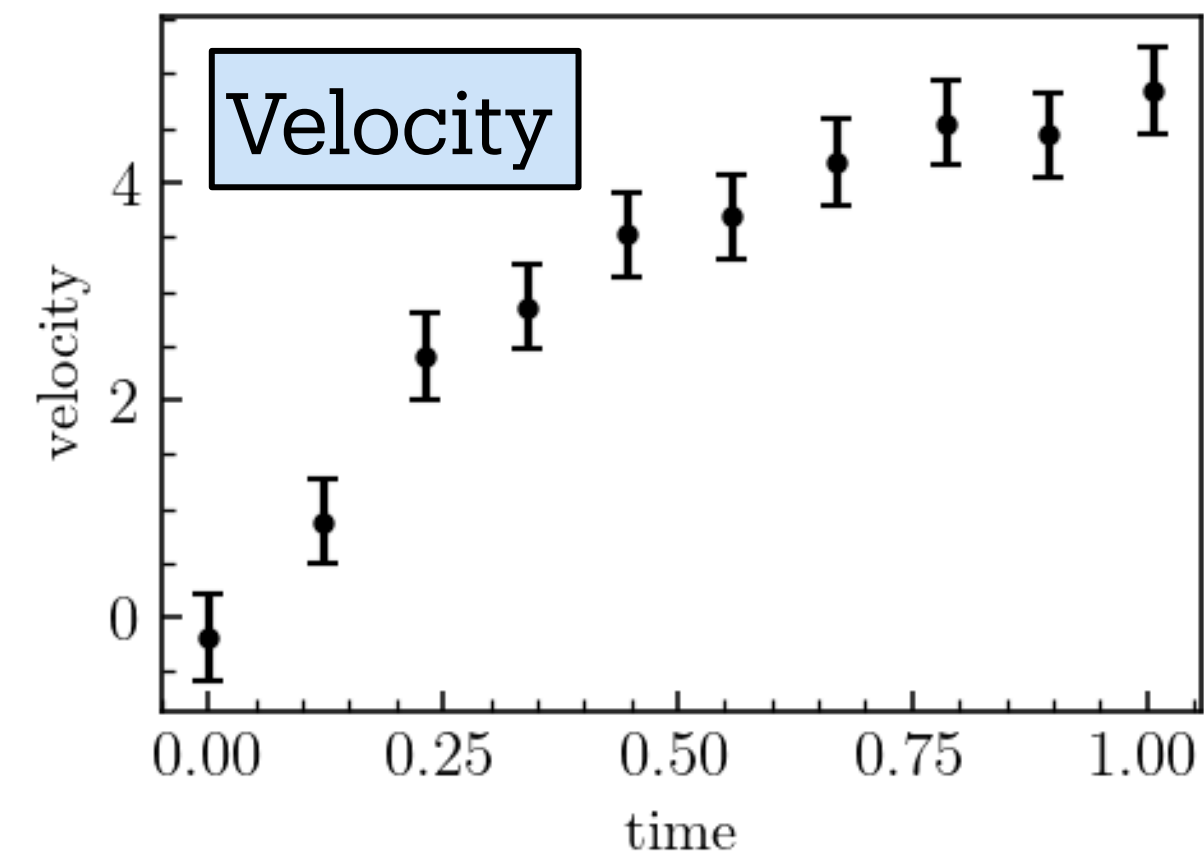
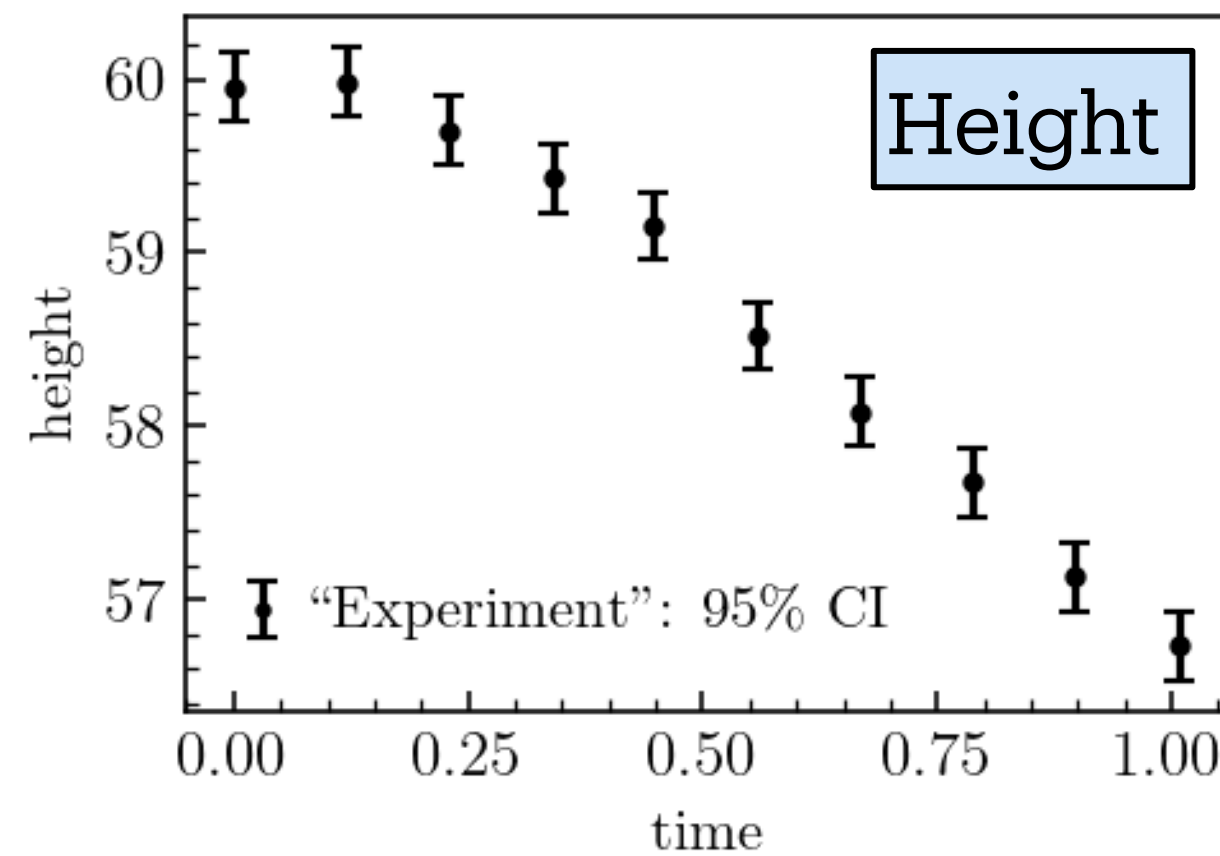
EoM:

$$a = \frac{d\mathbf{v}}{dt} = \mathbf{g} - 0.4v^2 \hat{\mathbf{v}}, \quad \frac{dh}{dt} = -v$$

$$h_0 = 0 \text{ m}, v_0 = 0 \text{ m/s}, \mathbf{g} = 9.8 \text{ m/s}^2$$

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Theoretical model (theory):

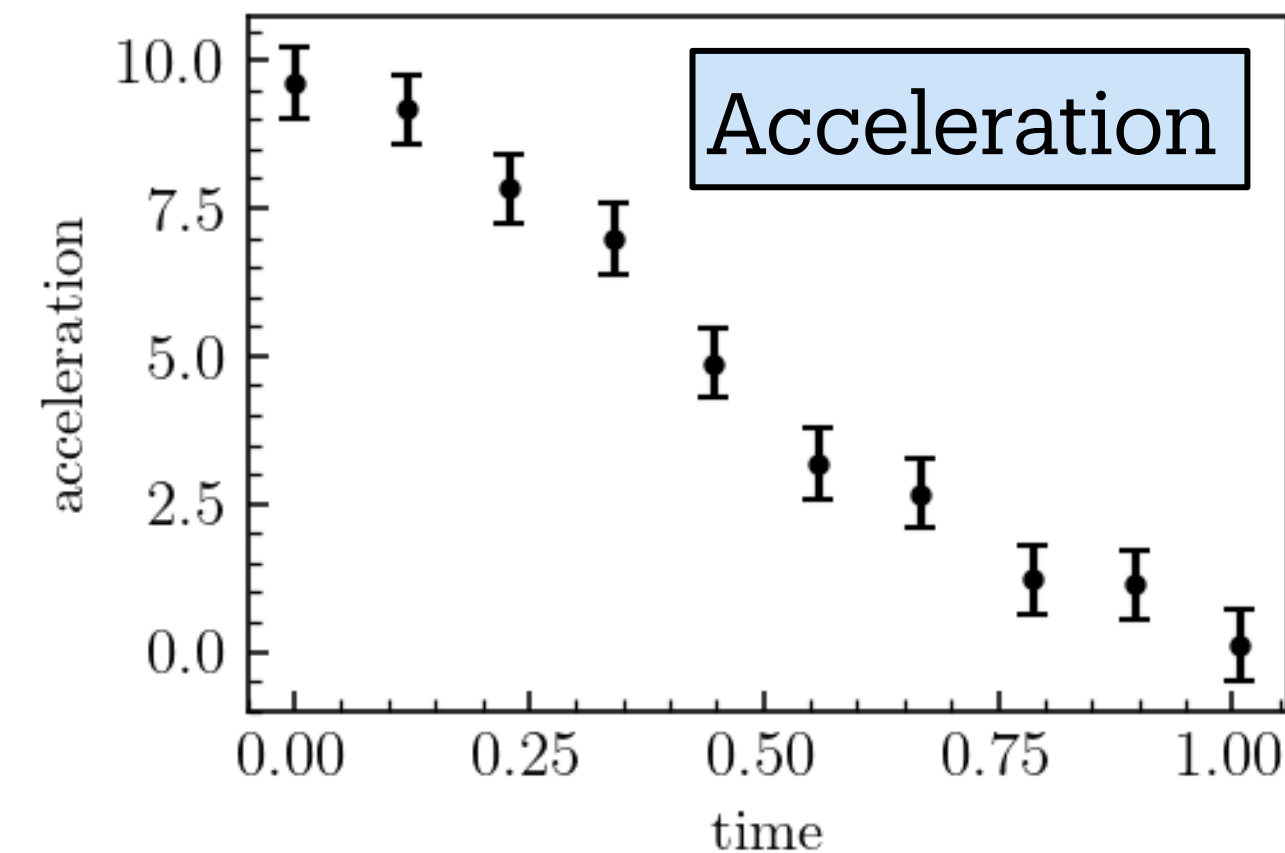
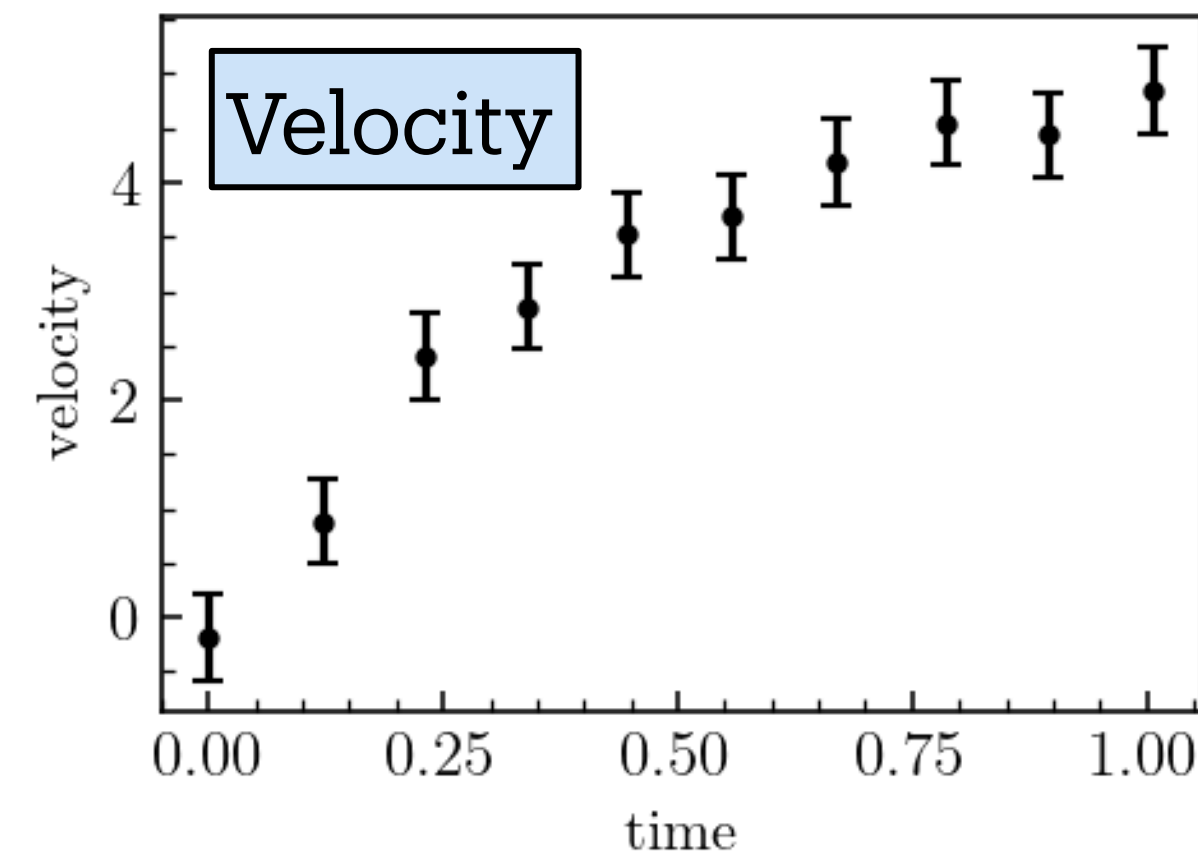
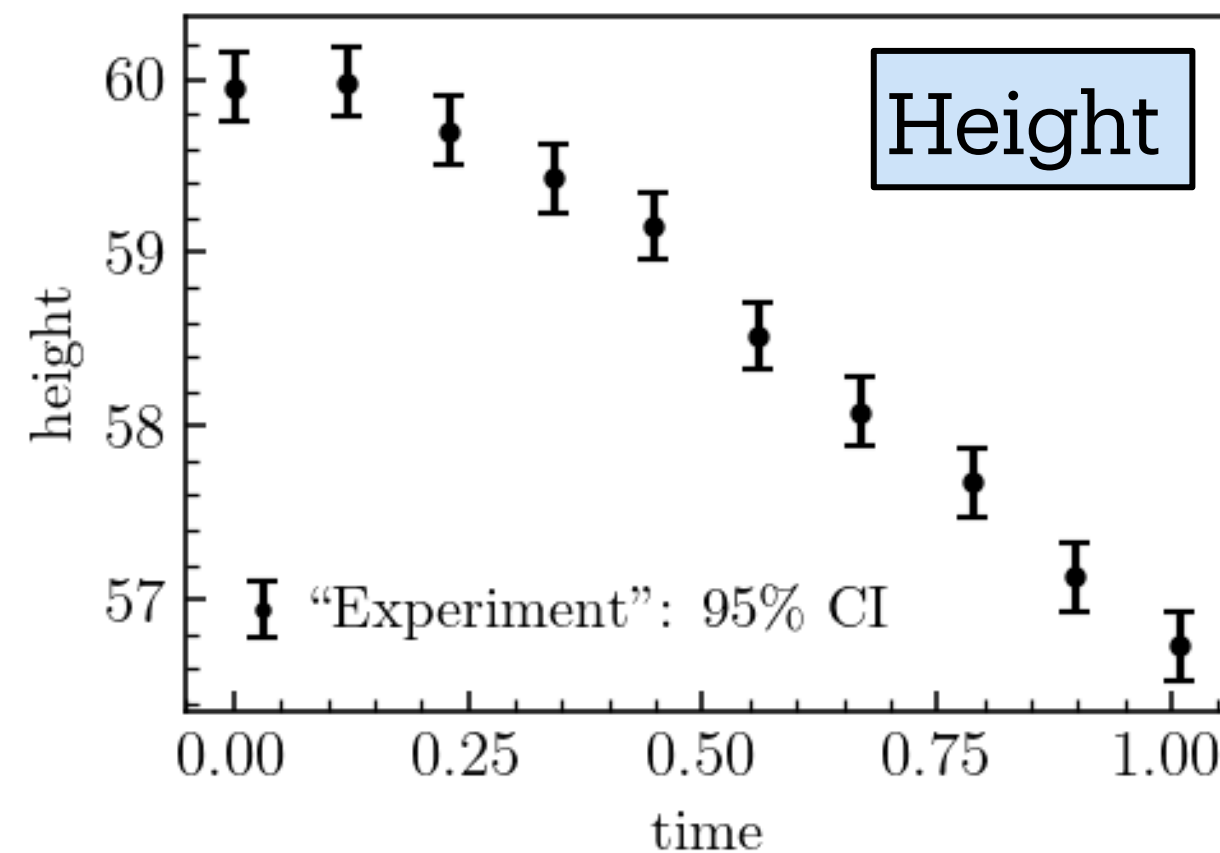
EoM: $a = g, \quad v = v_0 + gt, \quad h = h_0 - v_0 t - \frac{1}{2}gt^2$

2 physical model parameters : g, v_0

Goal: Infer the correct value $g = 9.8 \text{ m/s}^2$ from data (also v_0).

Motivation: A simple experiment

A ball is dropped from a tower. Height, velocity, acceleration are measured with time.



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 $a = \frac{dv}{dt} = g - 0.4v^2 \hat{\mathbf{v}}, \quad \frac{dh}{dt} = -v$

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Bayesian updating of knowledge

$$\text{pr}(A|B, I) = \frac{\text{pr}(B|A, I)\text{pr}(A|I)}{\text{pr}(B|I)} \Rightarrow \underbrace{\text{pr}(\alpha|\mathbf{y}_{\text{exp}}, I)}_{\text{posterior}} \propto \underbrace{\text{pr}(\mathbf{y}_{\text{exp}}|\alpha, I)}_{\text{likelihood}} \times \underbrace{\text{pr}(\alpha|I)}_{\text{prior}}$$

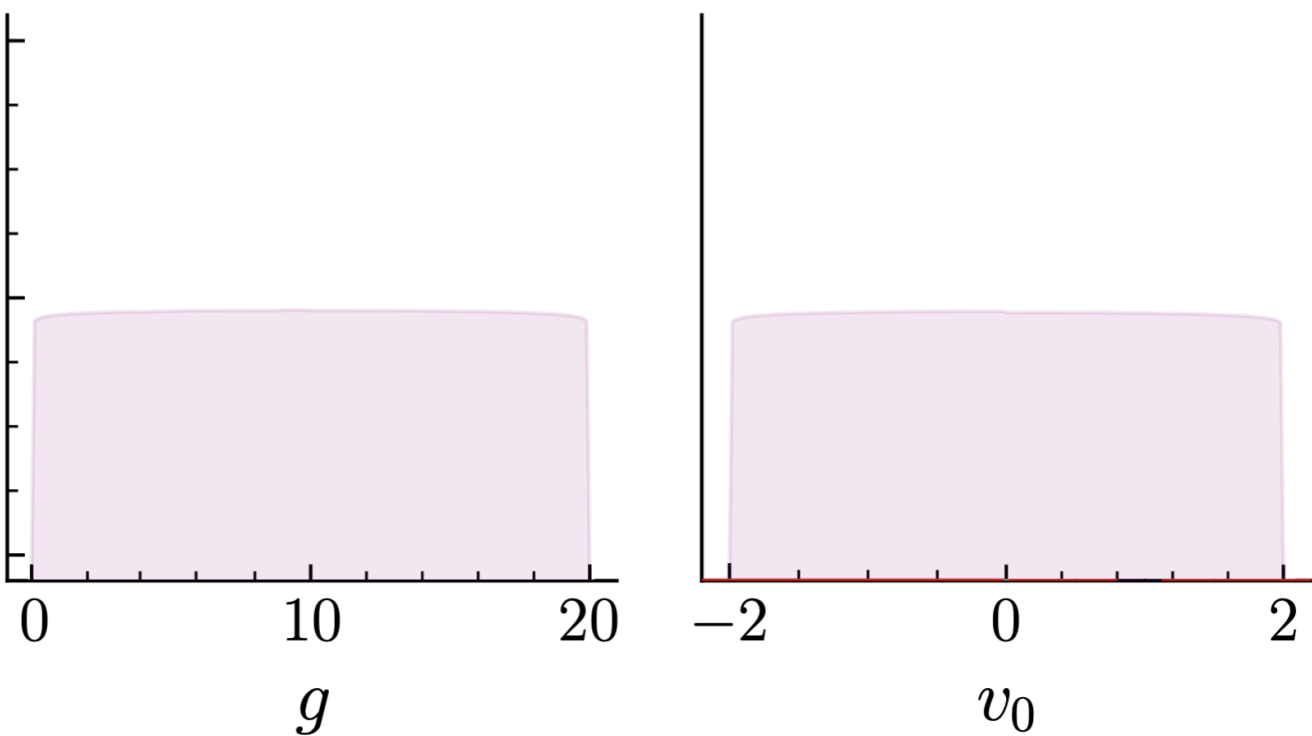
Continuing with the Ball-drop notebook - 1

- Only height is measured in the simplified ball-drop version in the notebook.
- Everyone has different random noise → compare results to see sensitivity to data errors
- **Run through B.3: sampling and the first corner plot, without model discrepancy. (Ask any questions about the code! Physics model; priors; model discrepancy; samplers)**
- *Discuss these questions with your neighbors:*
 - What are the priors? Do they make sense?
 - Why are there correlations between g and v_0 ?
 - Why do you think the results for the physical parameters g and v_0 are so far off?
 - **Run the next cell generating plots of the fit versus data and TRUTH. New ideas?**
 - What might you do to make those results more accurate (closer to TRUTH)?

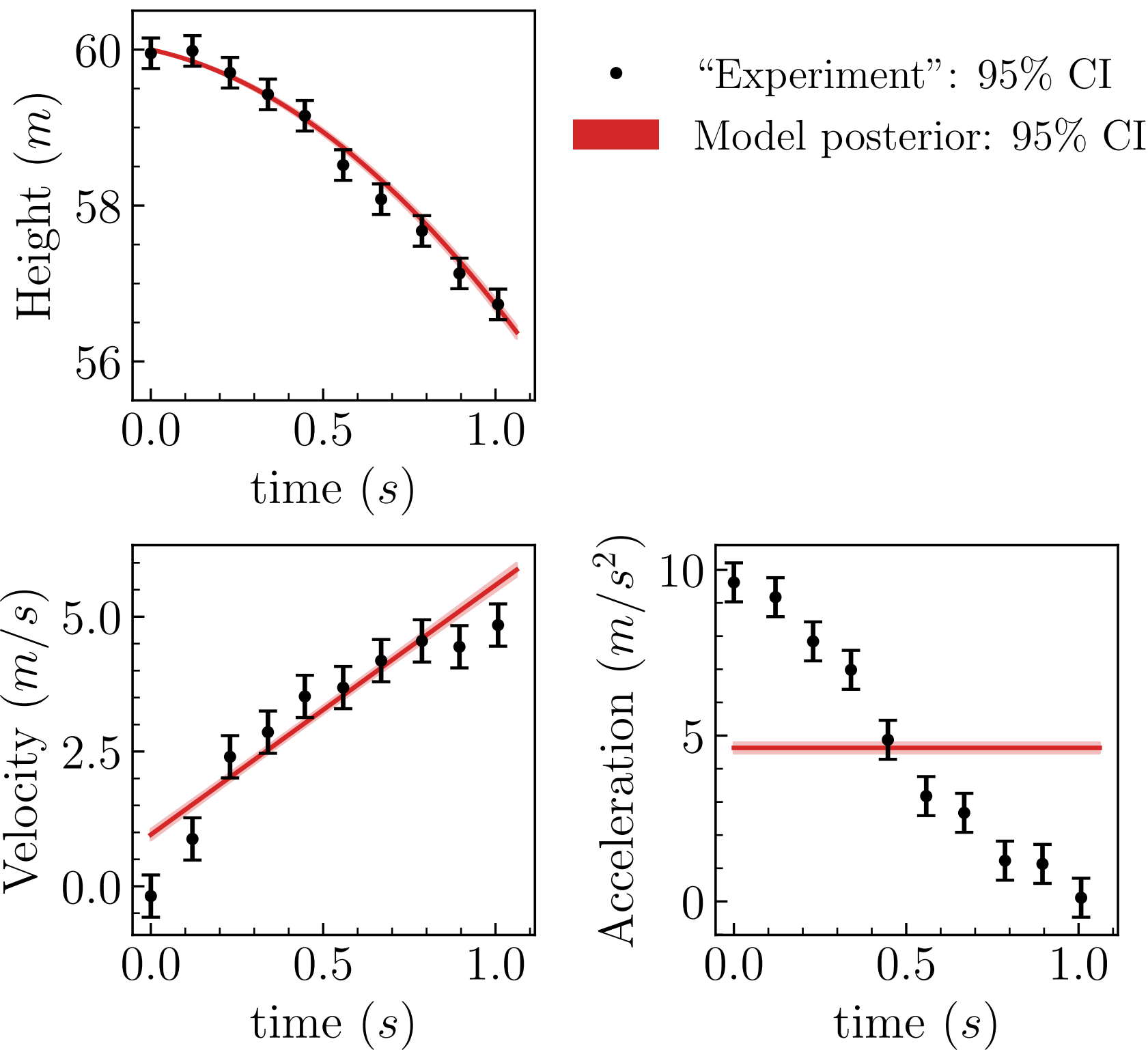
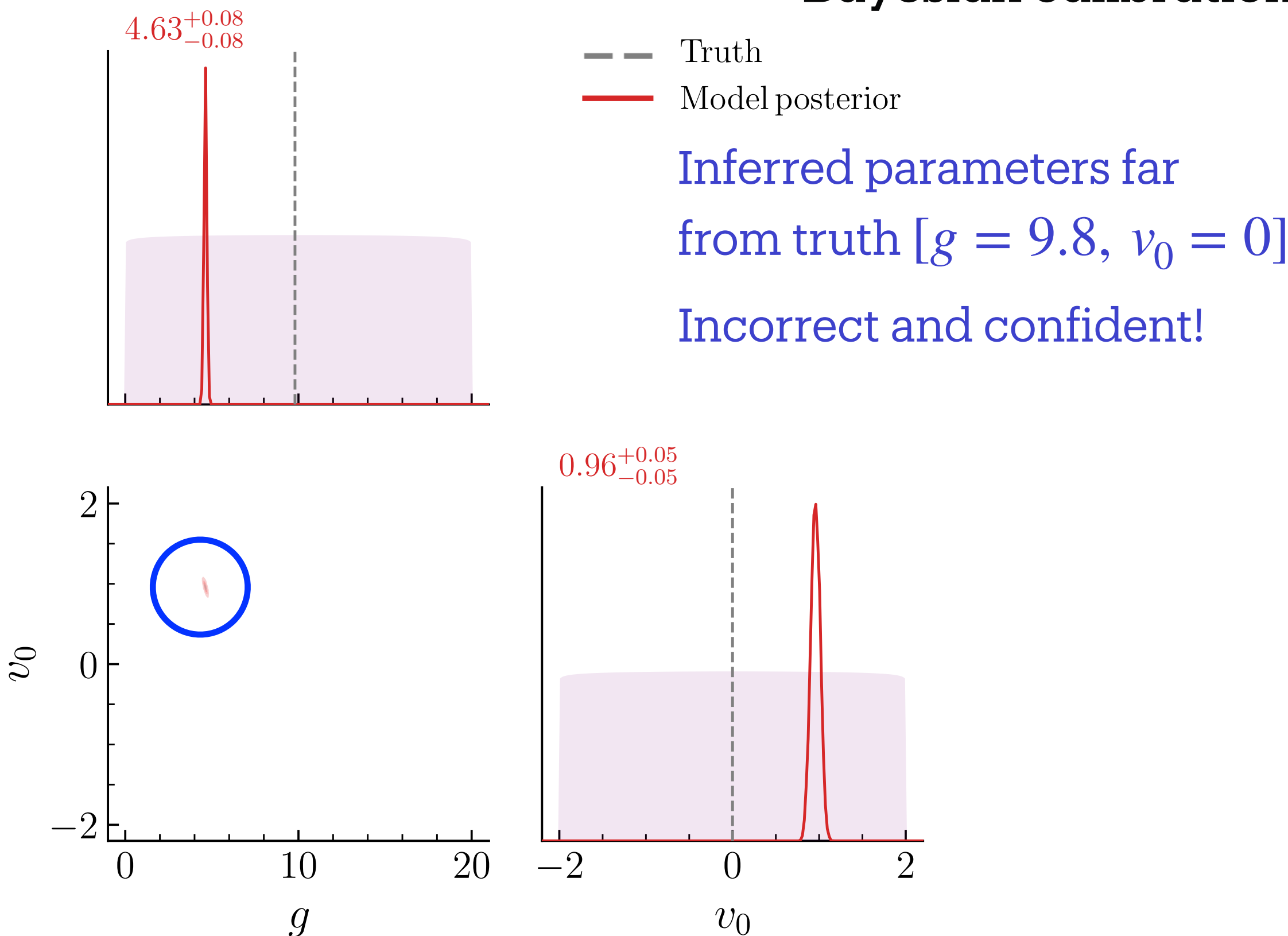
Model-data comparison: Bayesian inference

Prior information:

- The gravitational acceleration g is positive and < 20 (based on prior experiments and physical knowledge).
- The ball is released with an initial velocity close to 0 (information from the experimental setup).



Bayesian calibration with **all data**



Summary: What went wrong

- Reality has air drag. Theory does not capture this feature and tries to force the fit to data.
- Possible solution: Remove data at large times and re-fit. But where to draw the line?
- In general, all theories are approximations of true underlying physics; there will always be discrepancies at some level.
- **We need a method that correctly compares approximate theories with data, i.e., accounts for theoretical uncertainties arising from missing or approximate physics.**

Model discrepancy framework

$$\underbrace{y(t_i)}_{\text{Experimental observation}} = \underbrace{\eta(t_i, \theta)}_{\text{Predictions from approximate theory}} + \underbrace{\delta(t_i)}_{\text{Theoretical error}} + \underbrace{\epsilon(t_i)}_{\text{Observation error}}$$

$\zeta(t_i)$ Predictions from true theory

- Until now had set $\delta(t) \rightarrow 0$. Implicitly assuming theory is perfect. Parameters θ became **fitting variables**.
- **Prior information:** Theoretical model is not perfect. Observable prediction can be off even at true values of θ .
- $\delta(t)$ is an unknown function. Requires a functional prior (probability distribution over functions).
- Gaussian process provides a natural choice: $\delta(\cdot | \phi) \sim \text{GP}(\mathbf{0}, K(\cdot, \cdot | \phi))$

A Gaussian process is a probability distribution over functions...

M. Kennedy, A. O'Hagan
[doi:10.1111/1467-9868.00294](https://doi.org/10.1111/1467-9868.00294)

J. Brynjarsdóttir and A. O'Hagan,
[doi:10.1088/0266-5611/30/11/114007](https://doi.org/10.1088/0266-5611/30/11/114007)

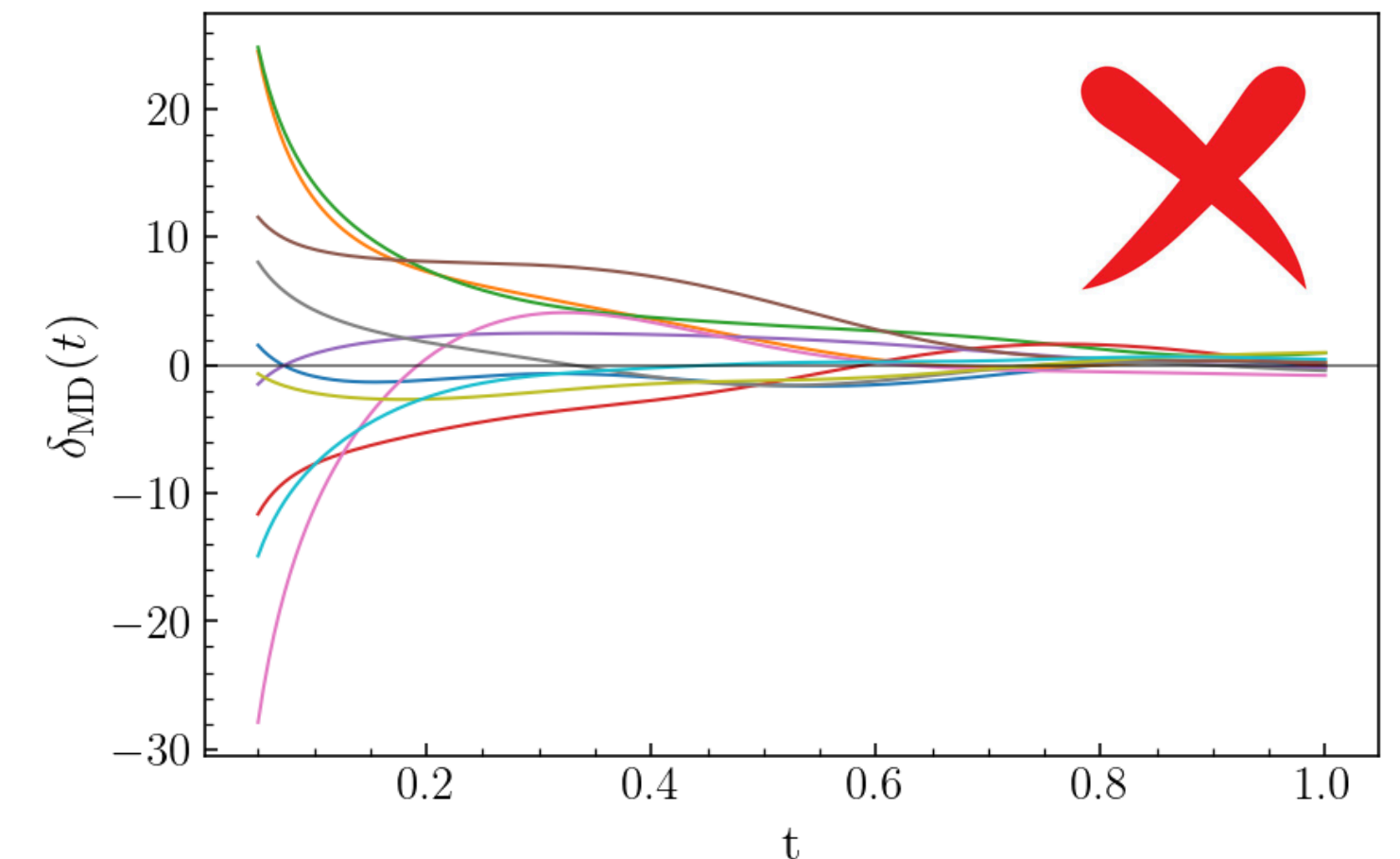
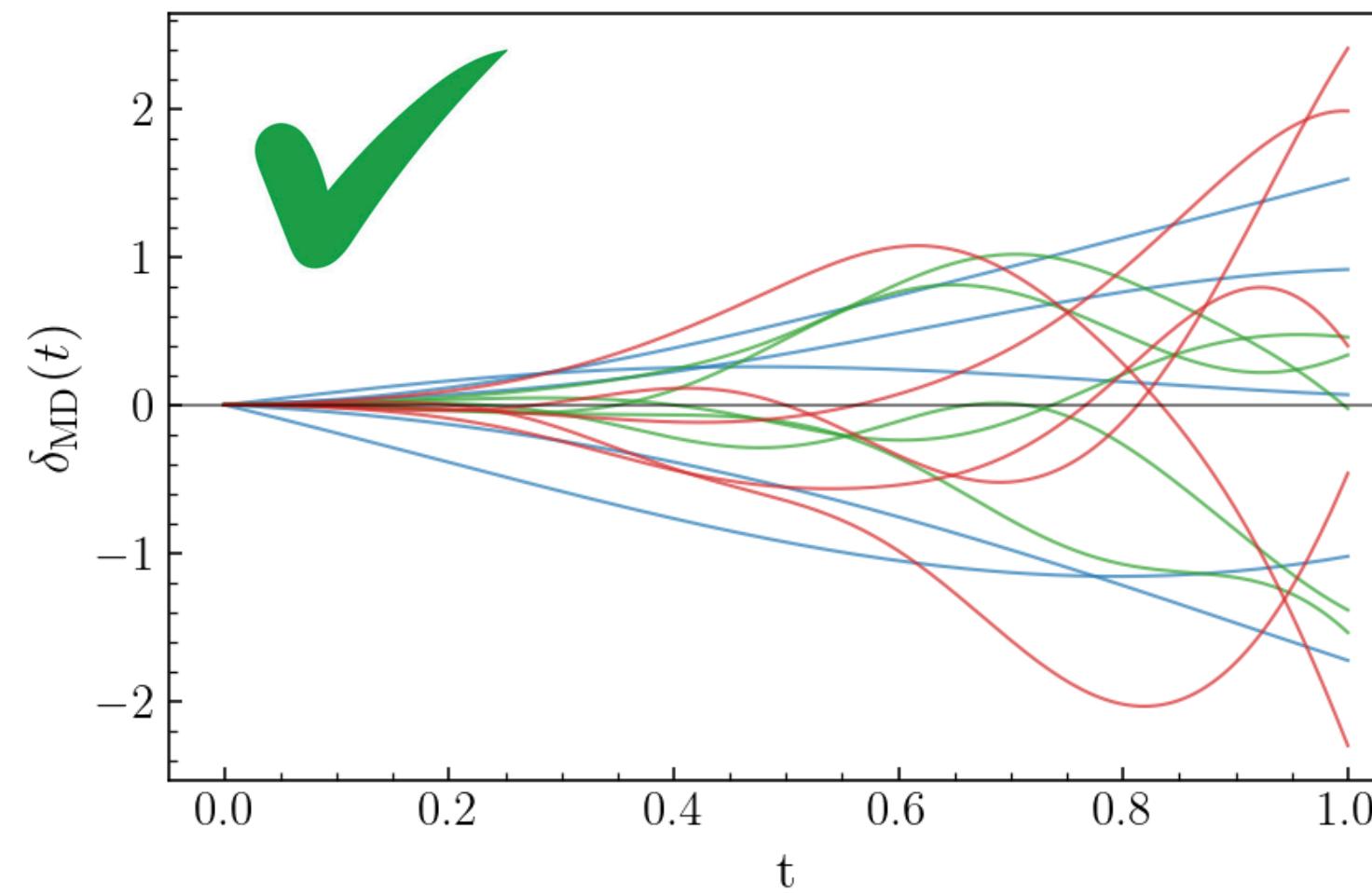
D. Higdon, M. Kennedy, et. al.,
[doi:10.1137/S1064827503426693](https://doi.org/10.1137/S1064827503426693)

Discrepancy function

- Prior information about limitations of theoretical model:
Air resistance has been neglected, and its effect is expected to become more important as the ball falls.
- $\delta(t)$ is then a **non-decreasing function**.
It may remain constant, including zero (if model is adequate over domain of experimental data), but otherwise its magnitude is allowed to grow as t increases.

$$\delta(\cdot | \phi) \sim \text{GP}(\mathbf{0}, K(\cdot, \cdot | \phi))$$

Allowed functions:

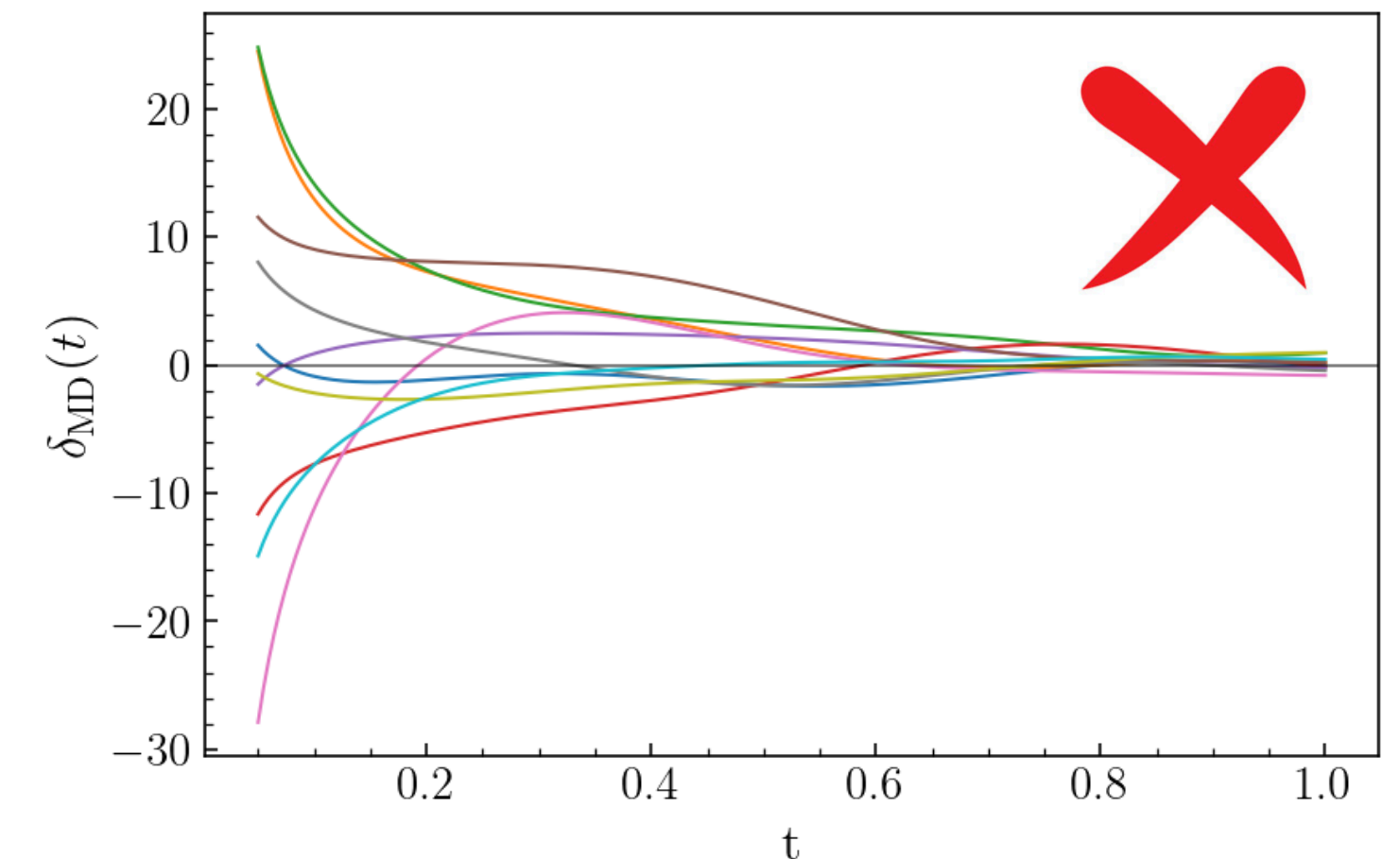
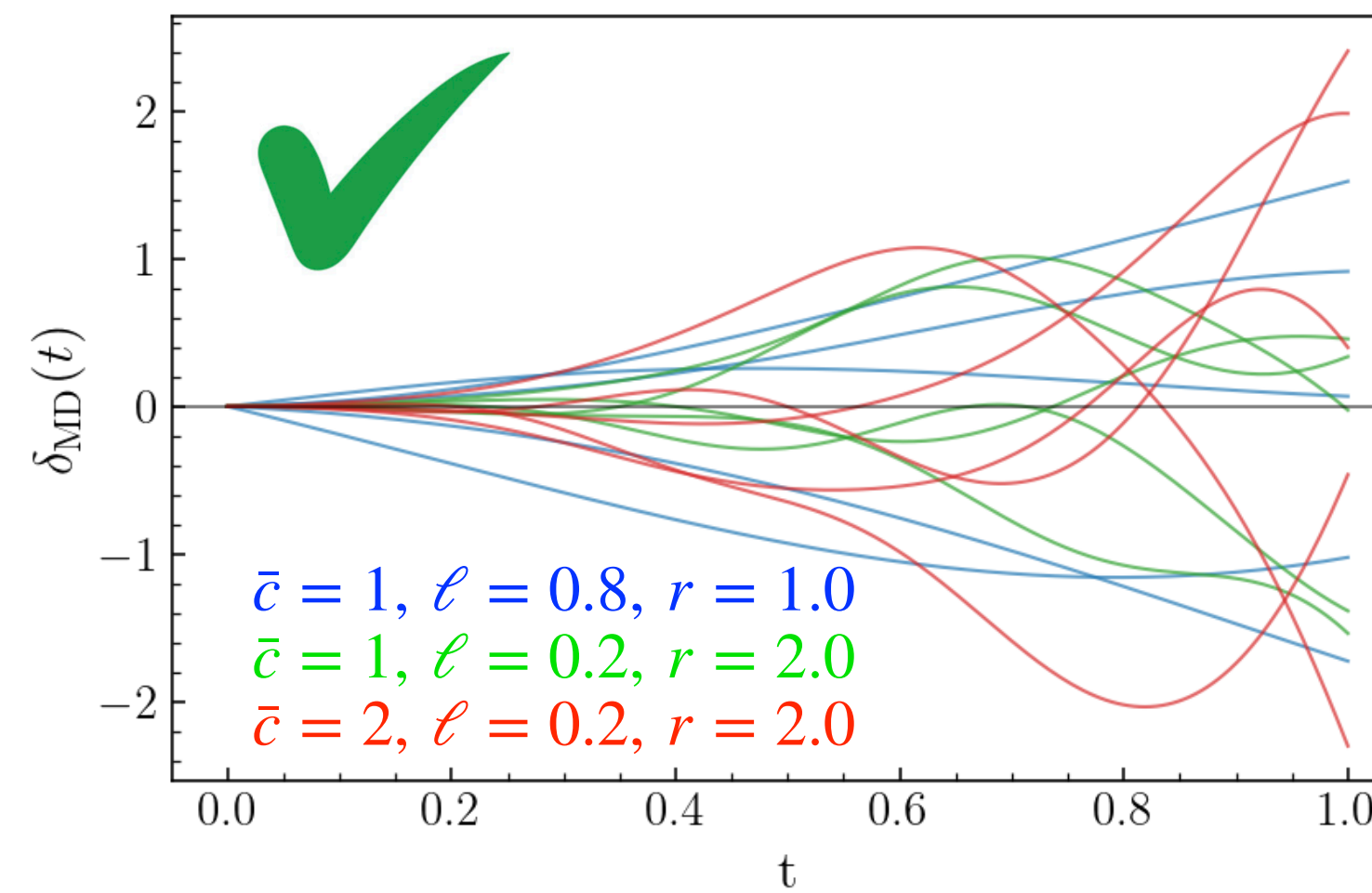


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Air resistance has been neglected, and its effect is expected to become more important as the ball falls.
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$$\delta(\cdot | \phi) \sim \text{GP}(\mathbf{0}, K(\cdot, \cdot | \phi))$$

$$K(t_i, t_j | \phi) \equiv \bar{c}^2(t_i t_j)^r \exp\left(-\frac{\|t_i - t_j\|^2}{2\ell^2}\right). \quad r \geq 0$$



GP kernel for such a functional prior

A Gaussian process is a probability distribution over functions that fit a set of points. The functions have special properties determined by the covariance kernel.

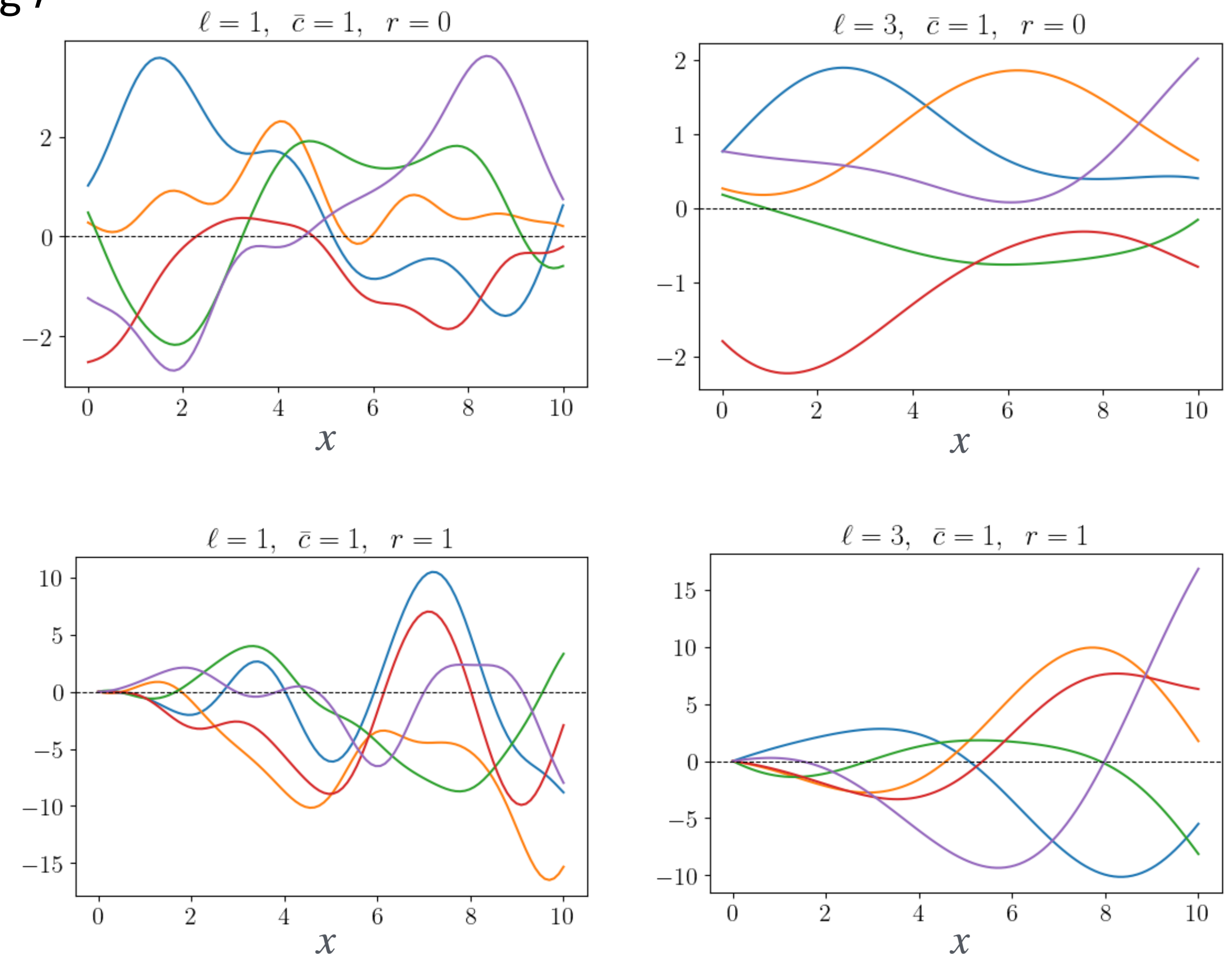
Consider the kernel:

$$K(x_i, x_j) = \bar{c}^2 (x_i x_j)^r \exp\left(-\frac{\|x_i - x_j\|^2}{2\ell^2}\right)$$

- The kernel, for fixed values of the parameters $(\bar{c}, r, \ell)$, generates a covariance matrix for a multivariate gaussian distribution.
- ℓ : correlation length between two points.
- $\bar{c}$: controls the overall scale over which the functions vary. As $\bar{c} \rightarrow 0$, GP predictions are Dirac-delta distributions.
- r (≥ 0): generates functions whose magnitude increase with x .

Increasing ℓ

Increasing r



Discrepancy kernel: two constructions

- Prior information about limitations of theoretical model:

1. Case I: Theory more correct at small t than at large t .
2. Case II: Theory more correct at small t than at large t **and correct at $t = 0$** .

- Corresponding kernels: [SJ, C. Shen, R. J. Furnstahl, U. Heinz, and M. T. Pratola, PLB 870, 139946 \(2025\), arXiv: 2504.13144](#)

➡ Kernel I: $K(t_i, t_j \mid \phi) \equiv s^2 + \bar{c}^2 (t_i t_j)^r \exp\left(-\frac{\|t_i - t_j\|^2}{2\ell^2}\right)$.

Encodes the knowledge (conservative):
Theory is more reliable at small t than at large t .

➡ Kernel II: $K(t_i, t_j \mid \phi) \equiv \bar{c}^2 (t_i t_j)^r \exp\left(-\frac{\|t_i - t_j\|^2}{2\ell^2}\right)$.

Encodes the knowledge (more informative):
Theory is correct at $t = 0$, but our confidence in its validity decreases as t increases

➡ Allows theory errors to be constant or increase with time (but not decrease).

➡ Non-vanishing theory error allowed for Kernel I ($s \neq 0$).

➡ $s = \bar{c} = 0 \implies \delta = 0$.

Bayesian inference with MD

$$\underbrace{y(t_i)}_{\text{Experimental observation}} = \underbrace{\eta(t_i, \theta)}_{\text{Theoretical prediction}} + \underbrace{\delta(t_i, \phi)}_{\text{Theoretical error}} + \underbrace{\epsilon(t_i)}_{\text{Observation error}}$$

Bayesian updating of knowledge

$$\text{pr}(A|B, I) = \frac{\text{pr}(B|A, I)\text{pr}(A|I)}{\text{pr}(B|I)} \implies \underbrace{\text{pr}(\alpha|\mathbf{y}_{\text{exp}}, I)}_{\text{posterior}} \propto \underbrace{\text{pr}(\mathbf{y}_{\text{exp}}|\alpha, I)}_{\text{likelihood}} \times \underbrace{\text{pr}(\alpha|I)}_{\text{prior}}$$

Fit $\zeta(t; \theta, \phi) \equiv \eta(t, \theta) + \delta(t, \phi)$ to data and estimate θ, ϕ simultaneously.

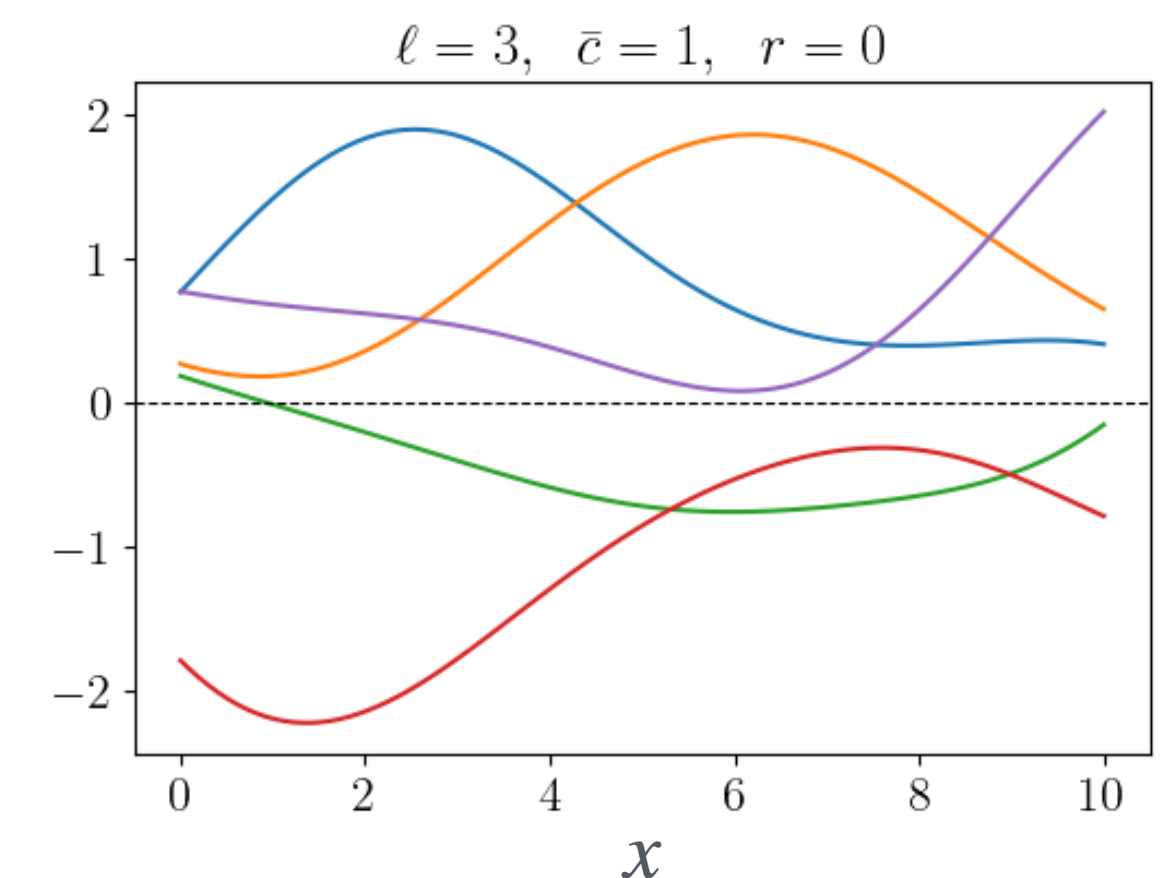
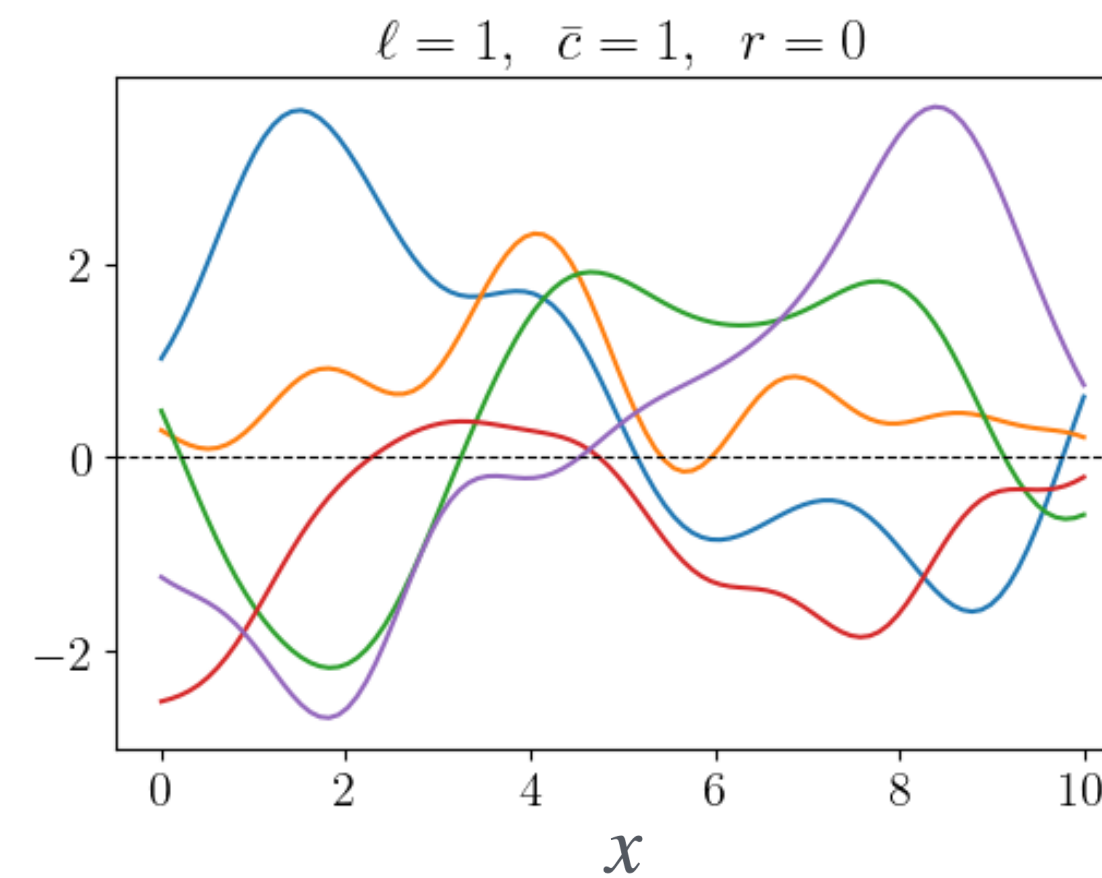
Note: individual discrepancy GP's (with hyperparameters) for each observable.

Beta priors mimicking uniform priors are considered for θ, ϕ .

Continuing with the Ball-drop notebook - 2

- First consider a *stationary* GP for the discrepancy

- Set $r = 0$
- Is this more or less informative than when $r \neq 0$?

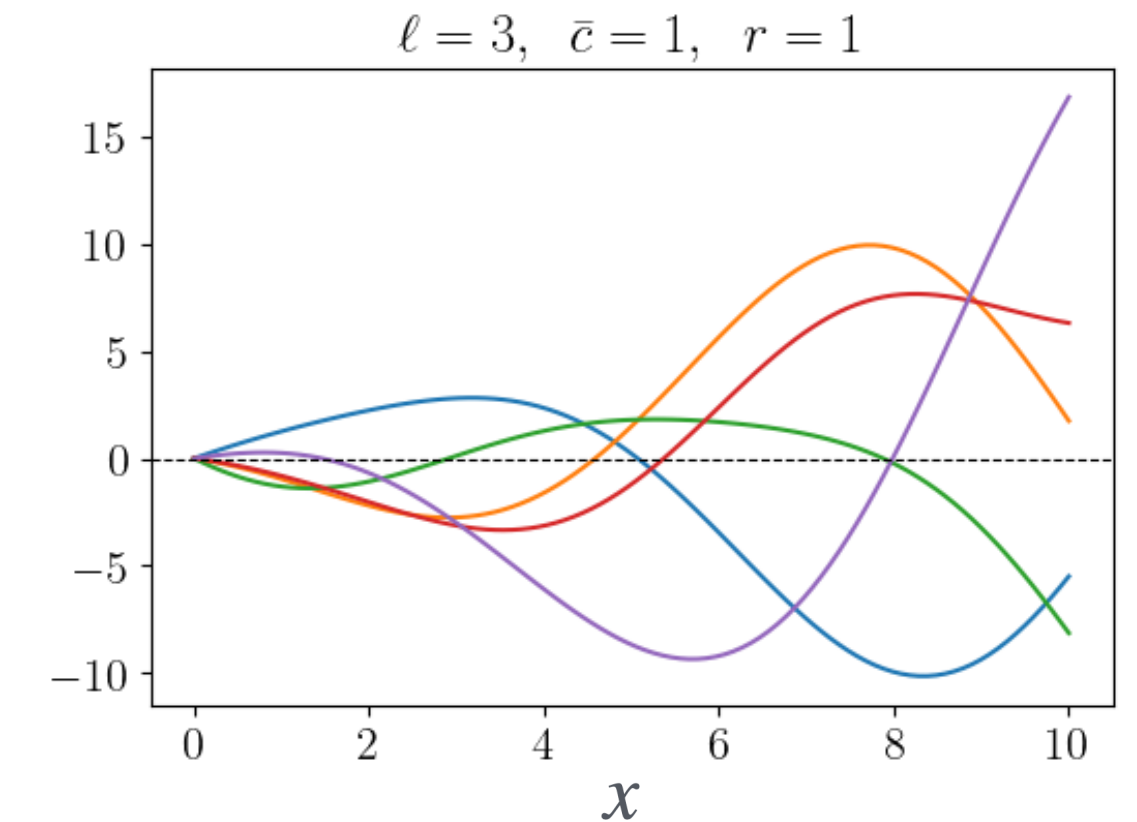
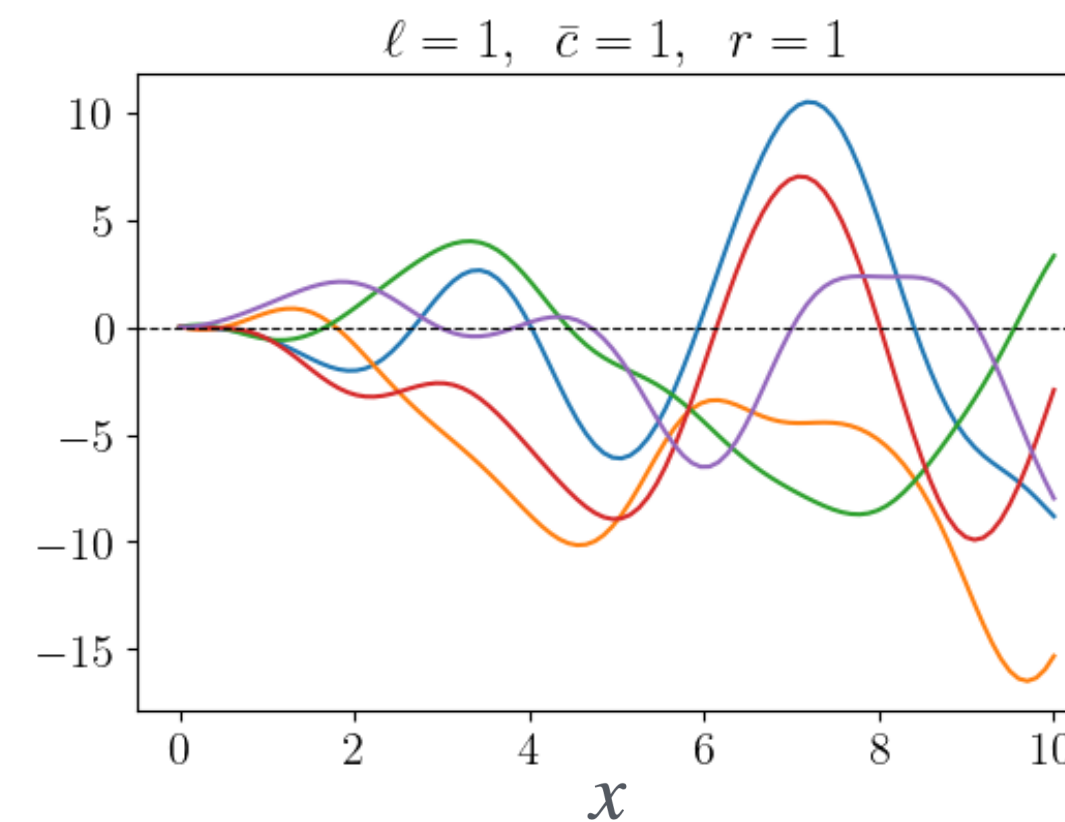


- Run through C1. and C2.: less informative discrepancy section**
- Discuss these questions with your neighbors:**
 - Do the results for g and v_0 come out better now? Consider both the mean and the variance.
 - What do the results for the hyperparameters $\bar{c}$ and ℓ tell us?

Continuing with the Ball-drop notebook - 3

- Now consider a **non-stationary** GP for the discrepancy: **Run through C3.**

- Sample r with prior $[0, 10]$
- What can you say about g, v_0 now?
- Do you think the sampling is ok?

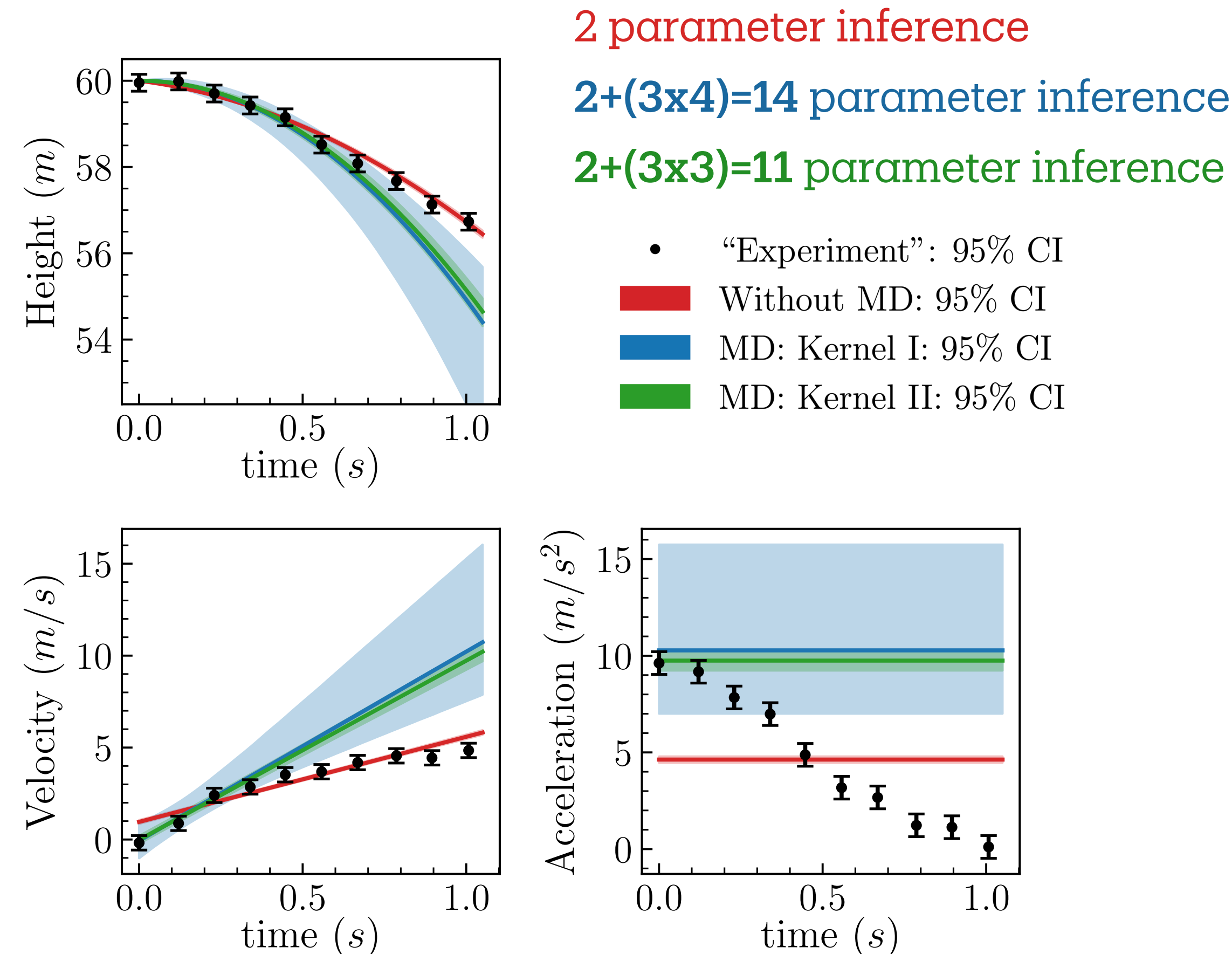
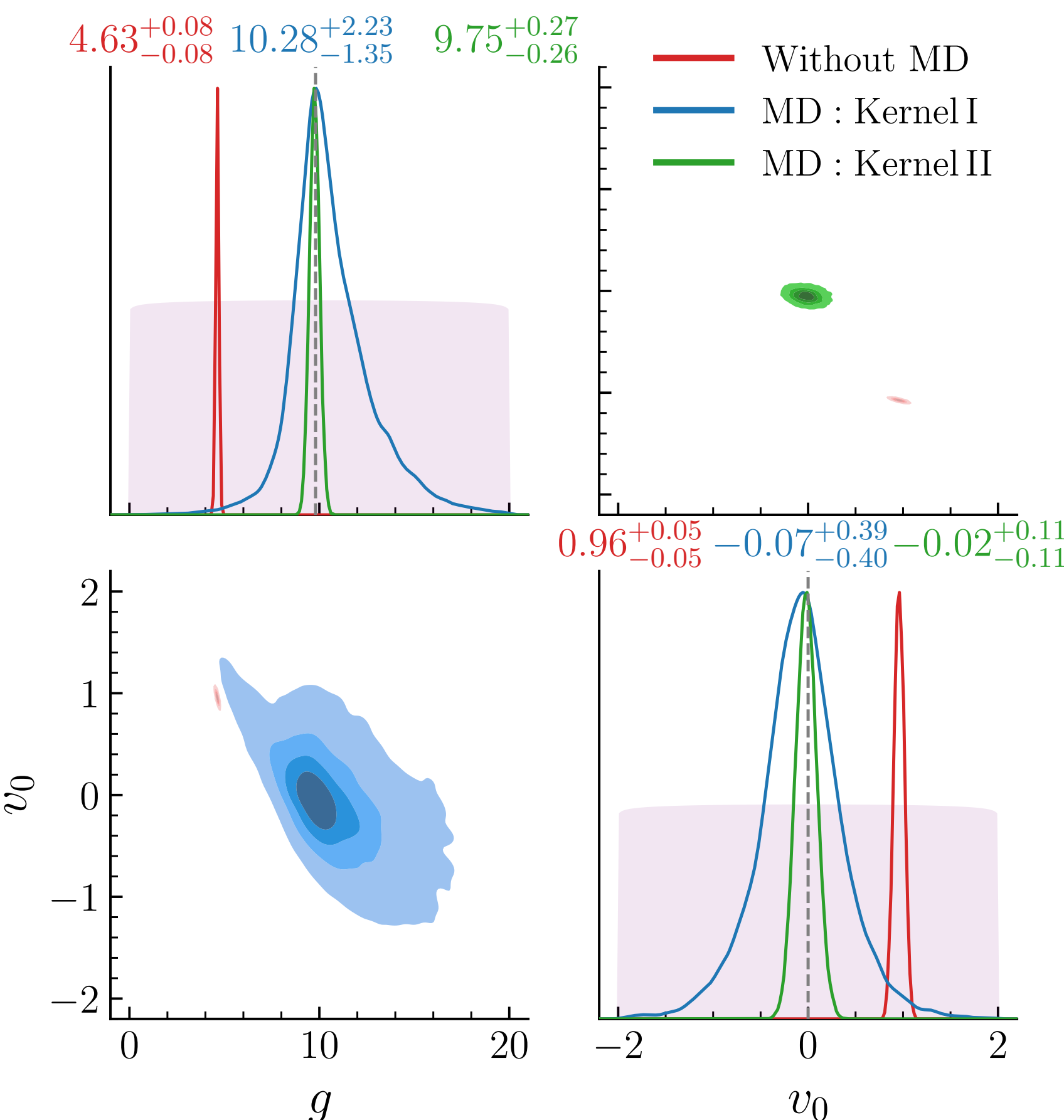


Go back to the tutorial README webpage

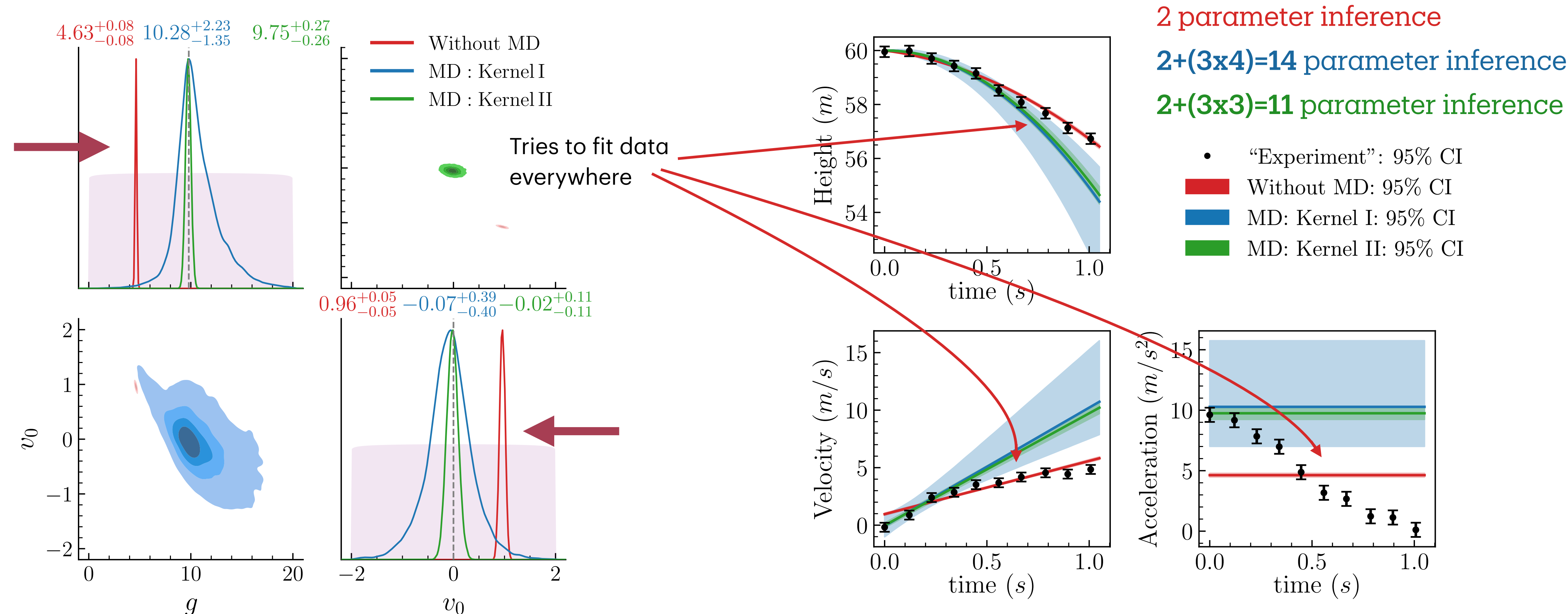
- Start a new Colab session with the ball-drop-plots-colab notebook (link at bottom)
- Do “run all” on the notebook (it will take some time generating pictures from pre-calculated results, which you can also reproduce on your own with even more time).

In the meantime, we'll discuss what you observed and generalize ...

Bayesian inference: Model predictions

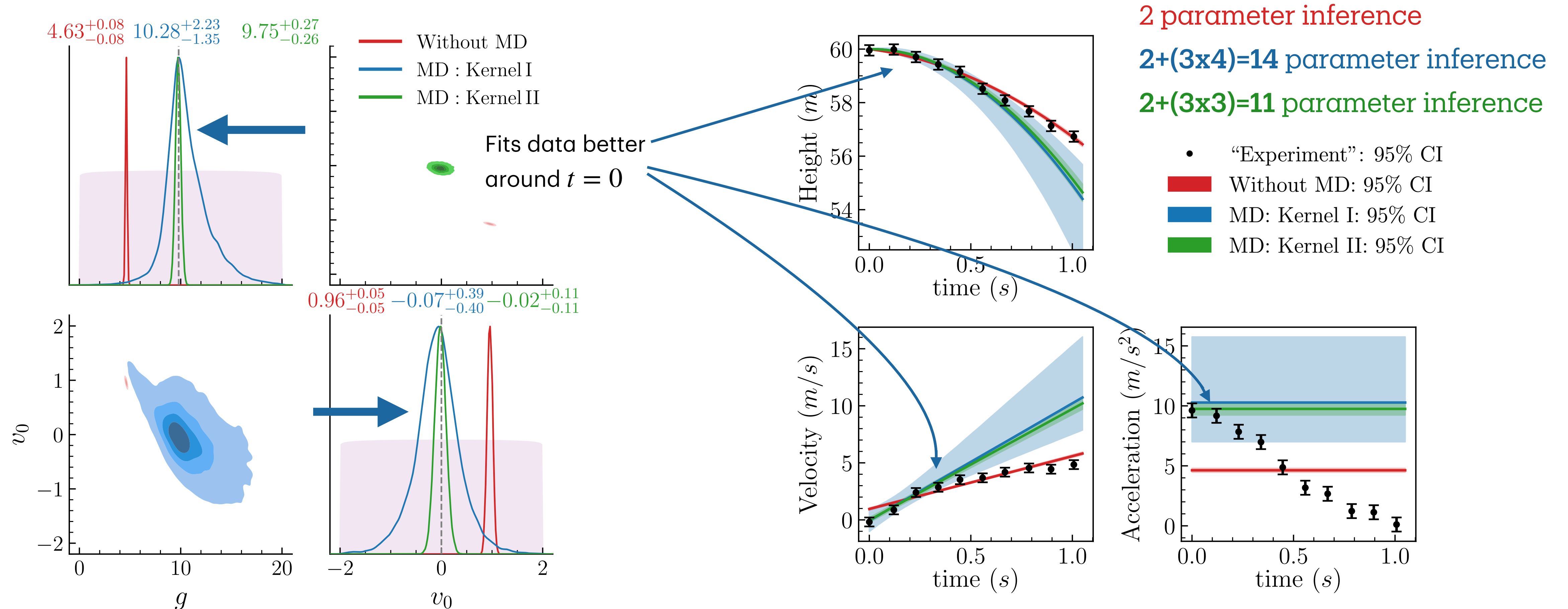


Bayesian inference: Model predictions



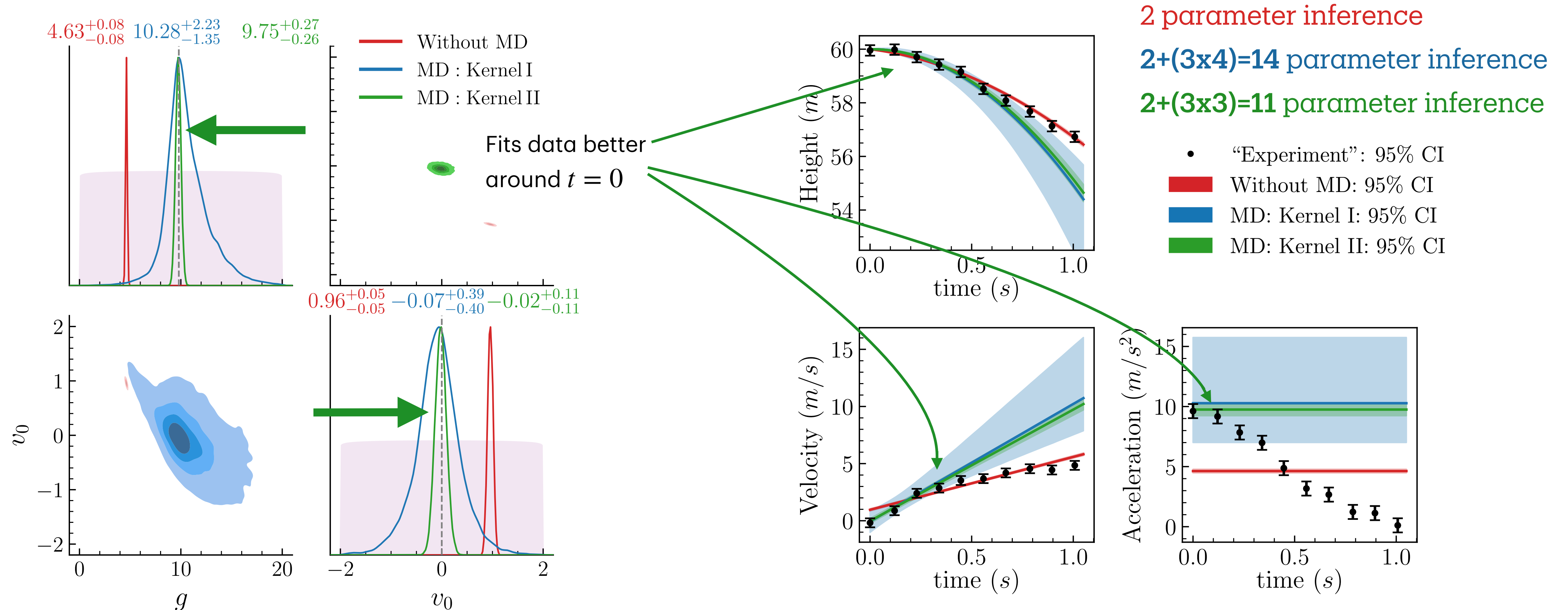
w/o MD (red): Inferred parameters far from truth [$g = 9.8$, $v_0 = 0$]; incorrect and confident.

Bayesian inference: Model predictions



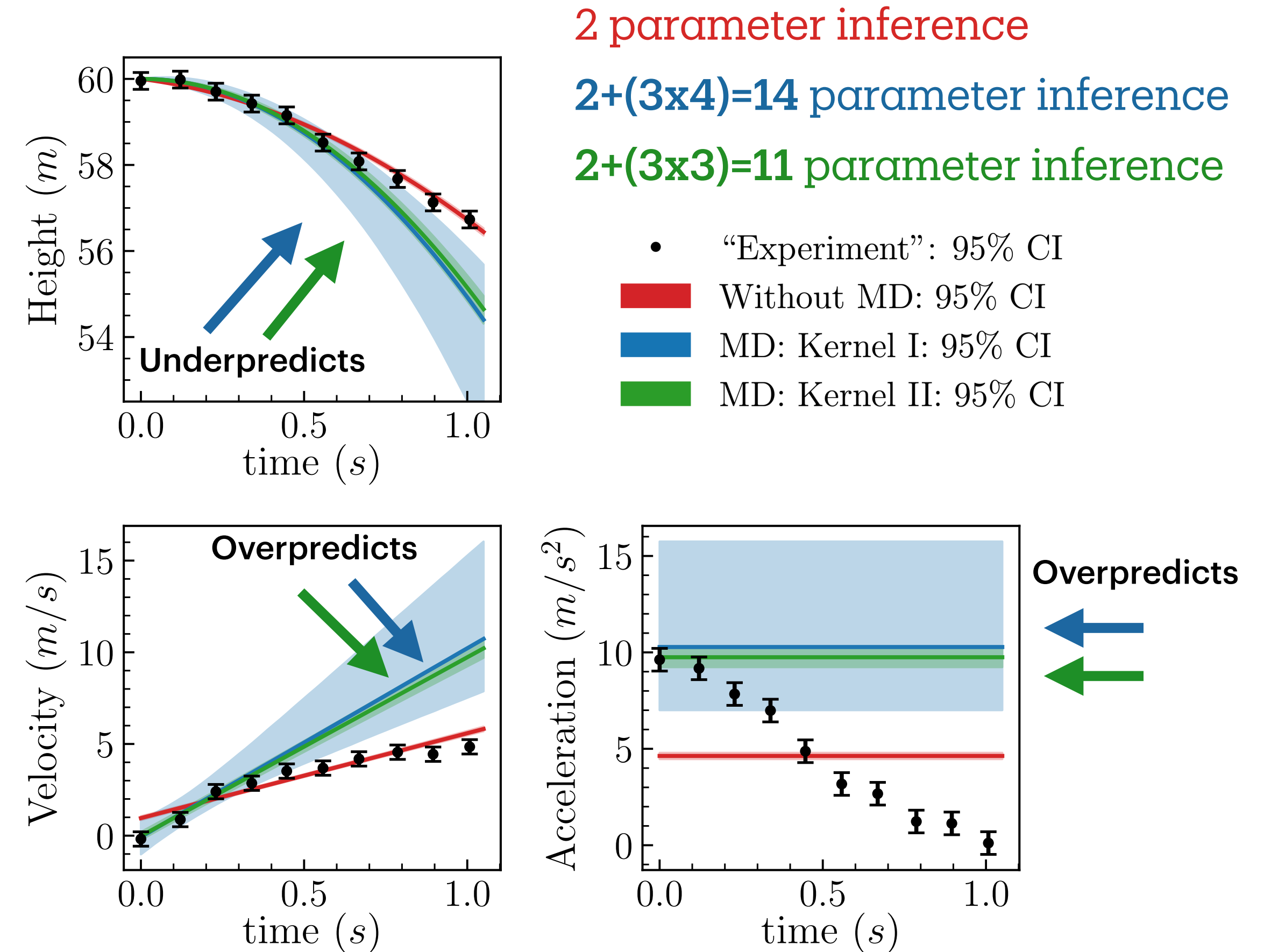
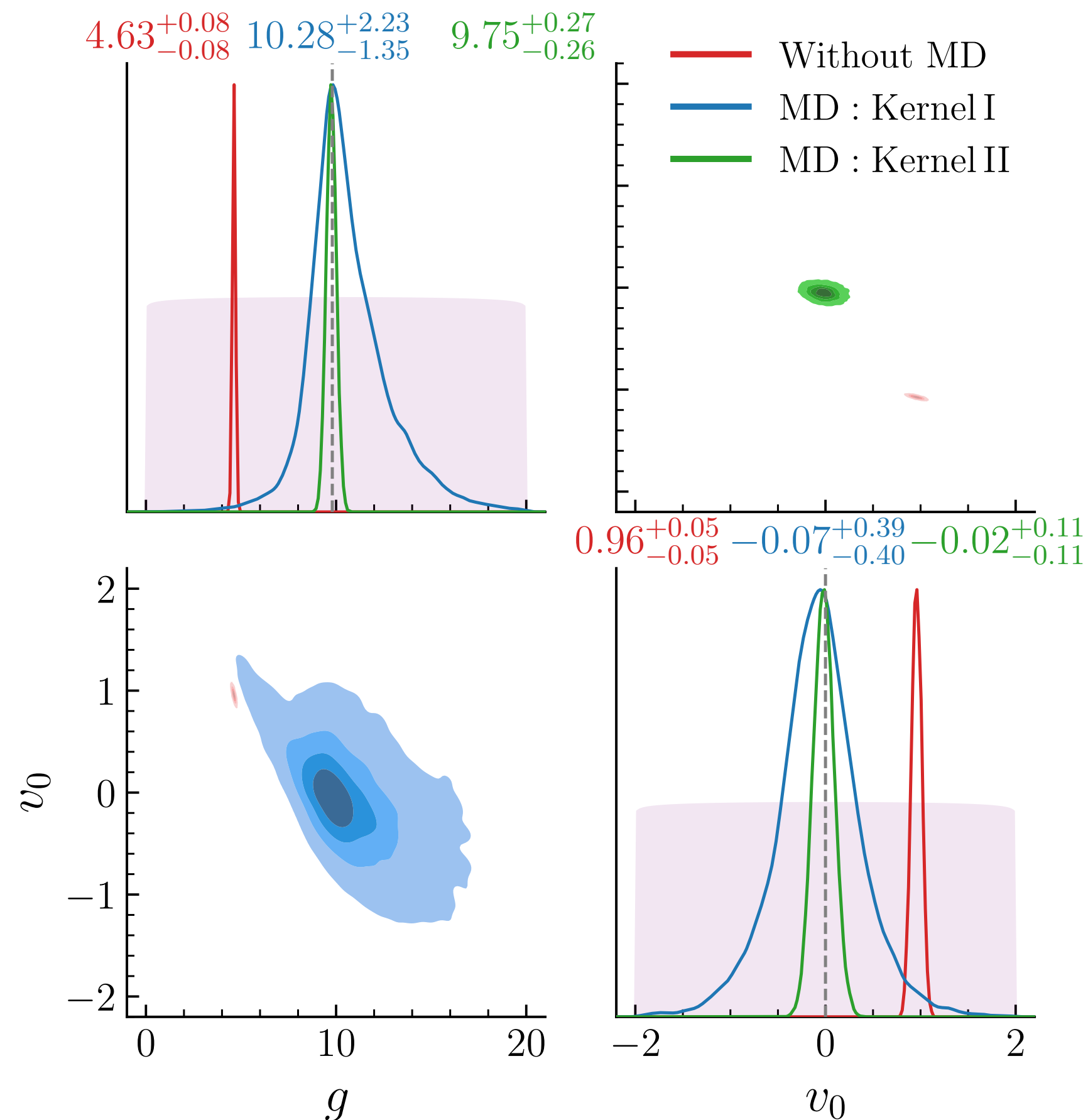
w/o MD (red): Inferred parameters far from truth [$g = 9.8$, $v_0 = 0$]; incorrect and confident.
w/ MD (blue — conservative kernel): g , v_0 true values inside posteriors.

Bayesian inference: Model predictions



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w/ MD (blue — conservative kernel): g , v_0 true values inside posteriors.
w/ MD (Green — more informative of theory error): g , v_0 posteriors correct and confident!

Bayesian inference: Model predictions



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w/ MD (blue — conservative kernel): g , v_0 true values inside posteriors.

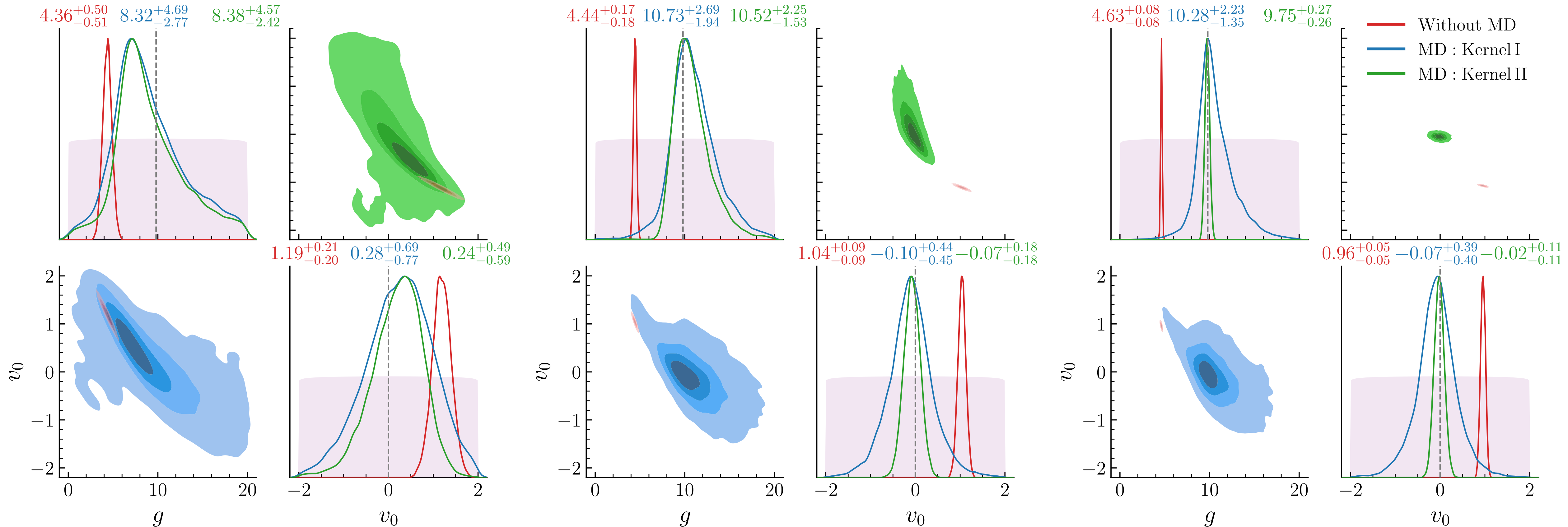
w/ MD (Green — more informative of theory error): g , v_0 posteriors correct and confident!

Check: stability of posteriors

Calibrate with **height**

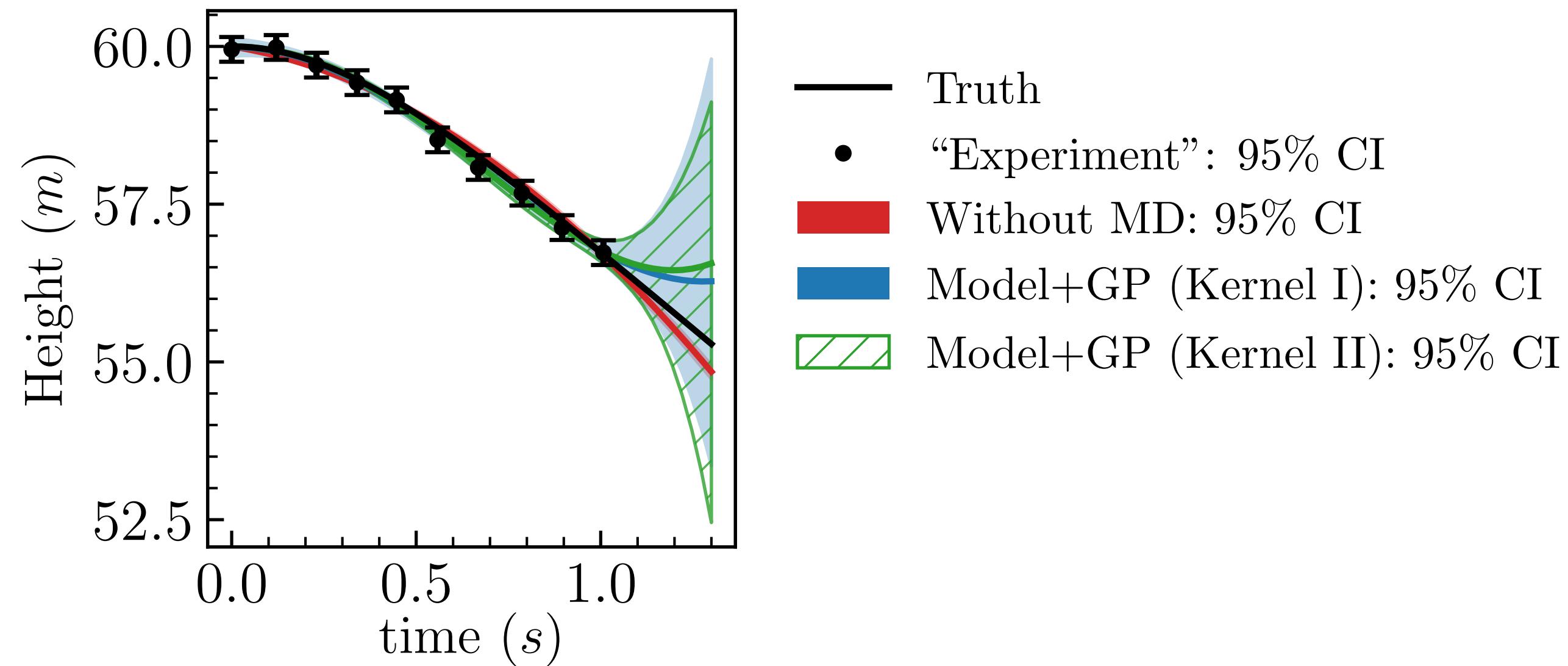
Calibrate with **height, velocity**

Calibrate with **all observables**

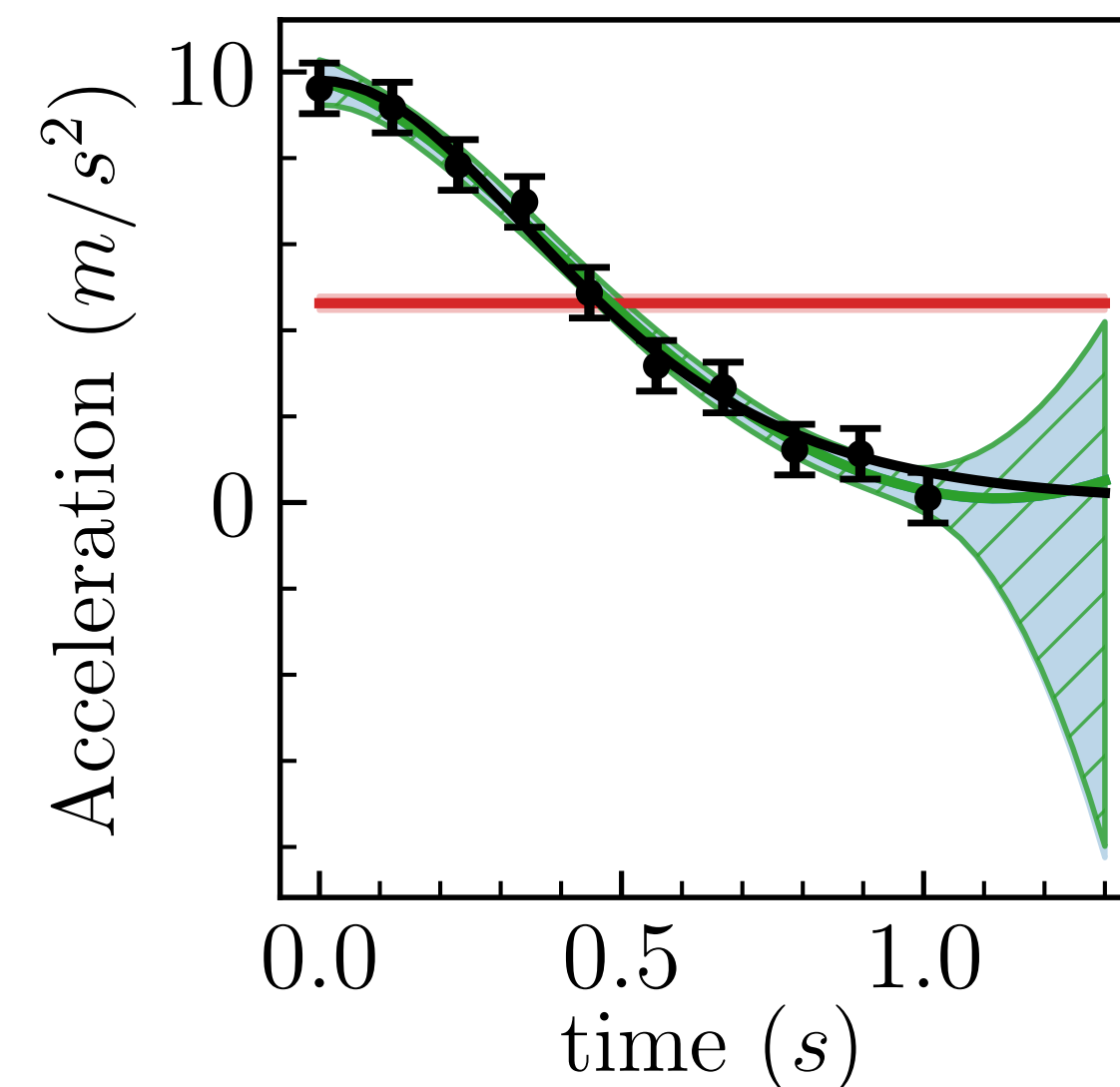
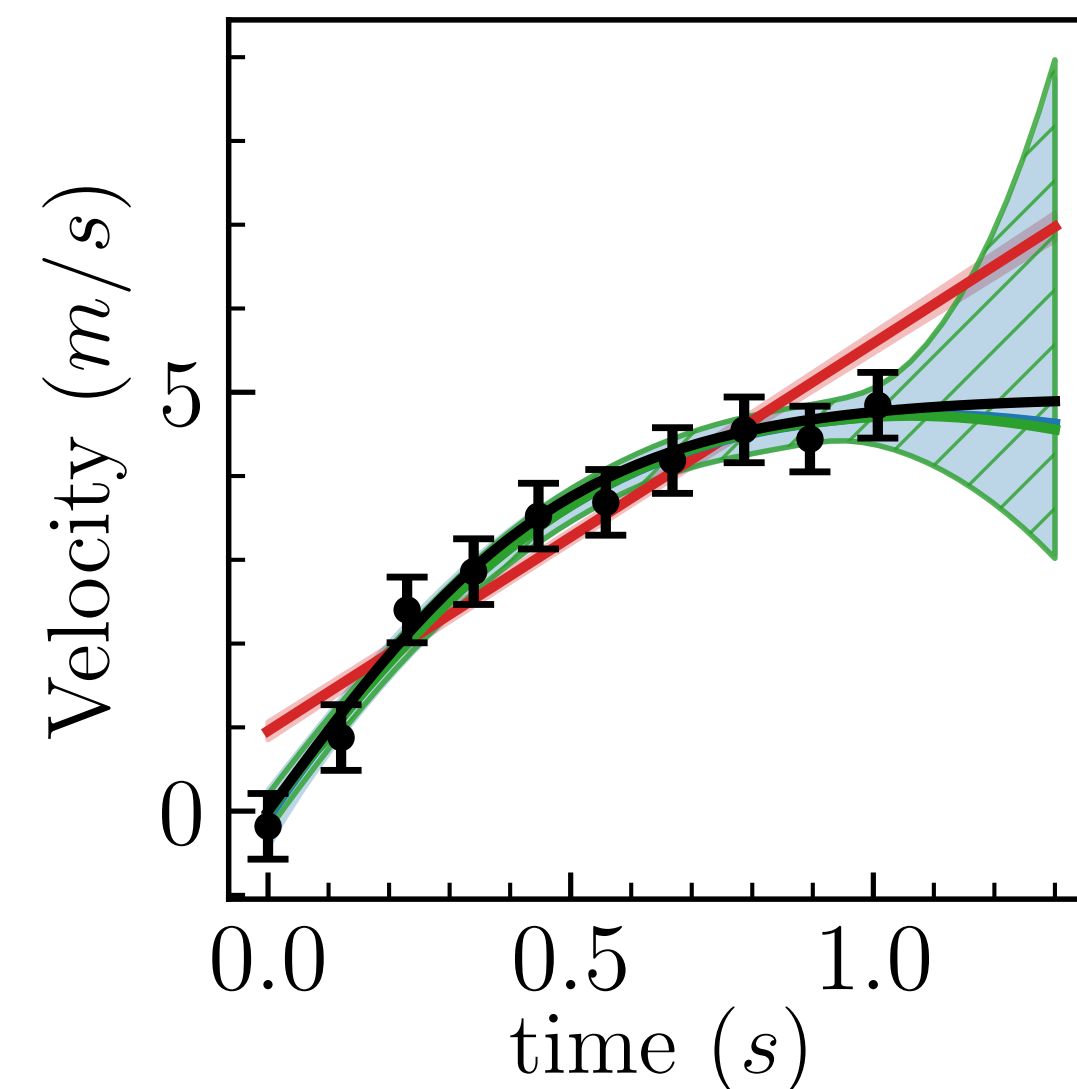


- w/o MD (red): Posteriors narrow around incorrect parameter value.
- w/ MD (blue, green): Posteriors narrow around correct parameter value.
More informative **Kernel II** posteriors within **Kernel I** (conservative kernel) — Consistent.

Bayesian inference: Model + discrepancy predictions



- Predictions from $\zeta(t; \theta, \phi) \equiv \eta(t, \theta) + \delta(t, \phi)$ *after updating the GP with the data*
- W/ MD predictions agrees with data.
- Extrapolation should be avoided.



→ Kernel I: $K(t_i, t_j | \phi) \equiv s^2 + \bar{c}^2(t_i t_j)^r \exp\left(-\frac{\|t_i - t_j\|^2}{2\ell^2}\right)$.

→ Kernel II: $K(t_i, t_j | \phi) \equiv \bar{c}^2(t_i t_j)^r \exp\left(-\frac{\|t_i - t_j\|^2}{2\ell^2}\right)$.

Plots from the DiscrepancyModel Ball-drop notebook

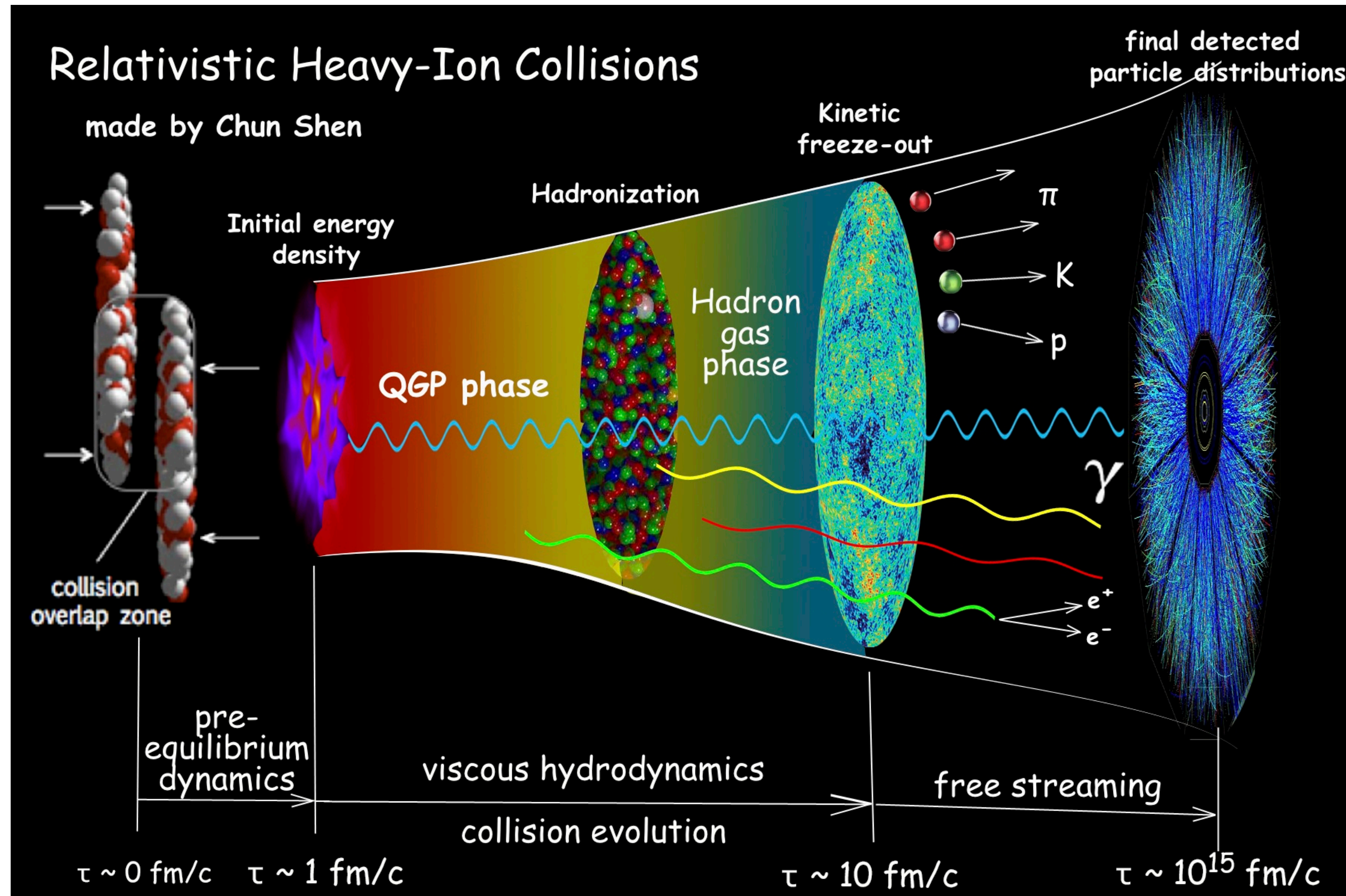
- Hopefully this has finished running in the background!
- There are instructions in the BAND DiscrepancyModel repository for running these locally (using a conda environment).
- What else do we learn from these plots?

Go back to the tutorial README webpage

- Start a new Colab session with the heavy-ion-plots-colab notebook
- Do “run all” on the notebook (it will take some time generating pictures from pre-calculated results, which you can also reproduce on your own with even more time).

In the meantime . . .

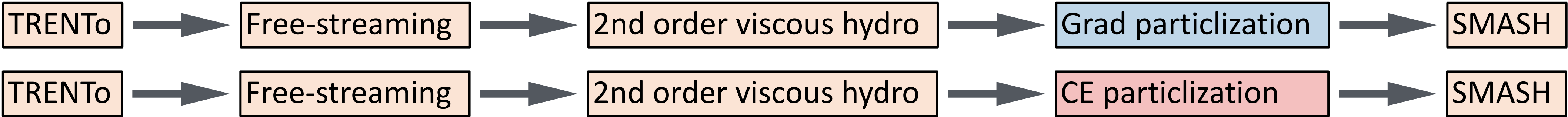
Heavy-ion collisions: Many stages of evolution



- Many stages. Many models. All models are approximate.
- Errors from one stage can propagate to other stages.

Bayesian parameter inference: Pb+Pb collision at 2.76 TeV

[SJ, PLB 874, 140243 (2026)]



$$\underbrace{y_{\text{exp}}(x)}_{\text{Experimental observation}} = \underbrace{\eta_{\text{mod}}(x, \theta)}_{\text{Predictions from approximate theory}} + \underbrace{\delta_{\text{MD}}(x)}_{\text{Error of approximate theory predictions}} + \underbrace{\epsilon(x)}_{\text{Observation error}}$$

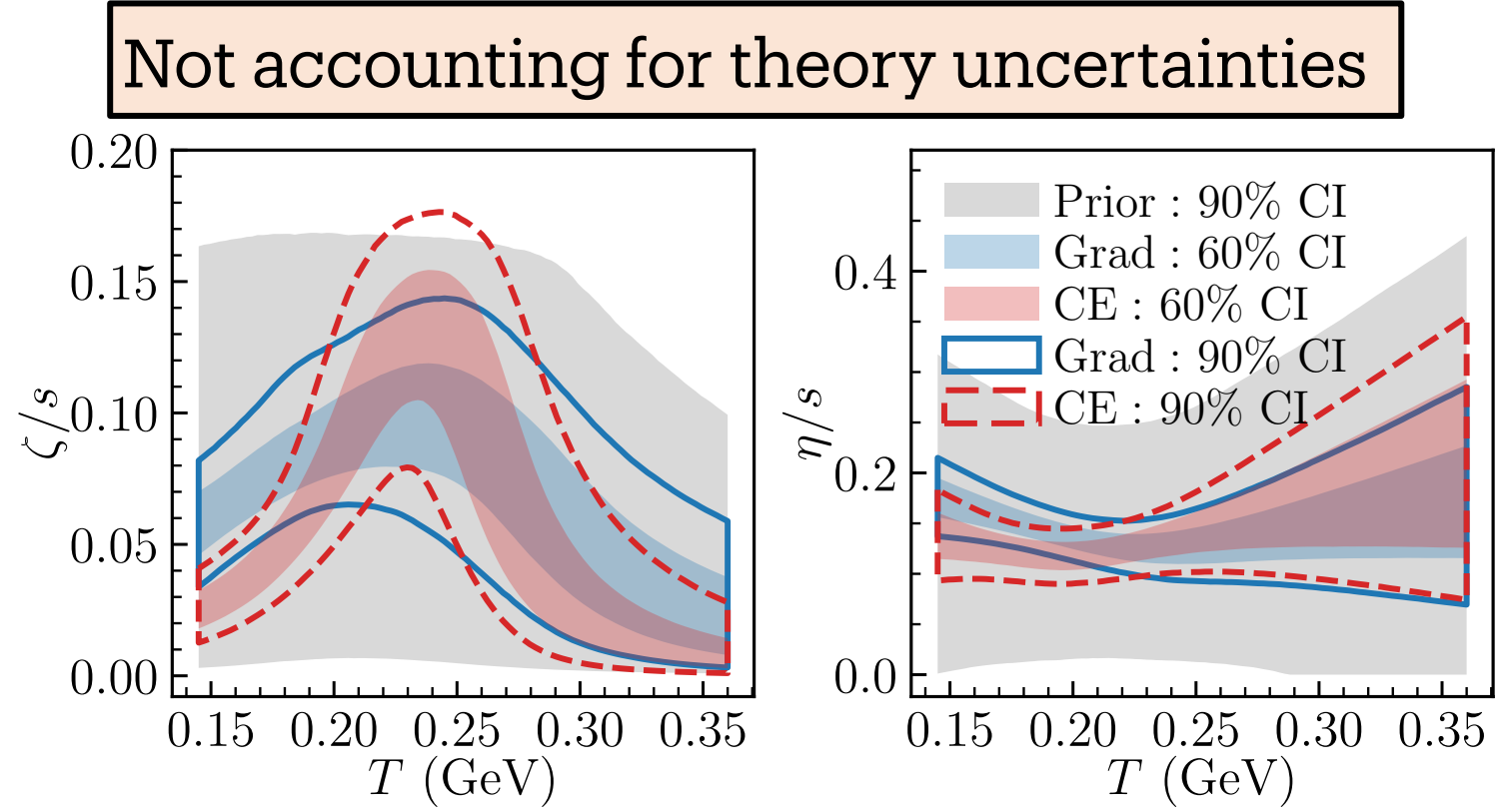
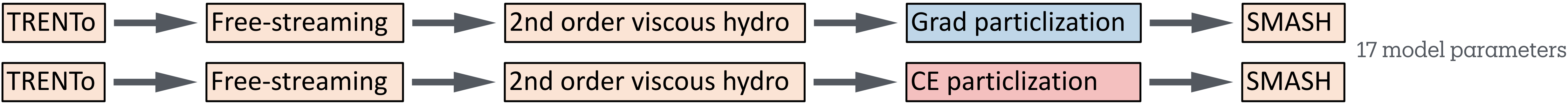
x : centrality

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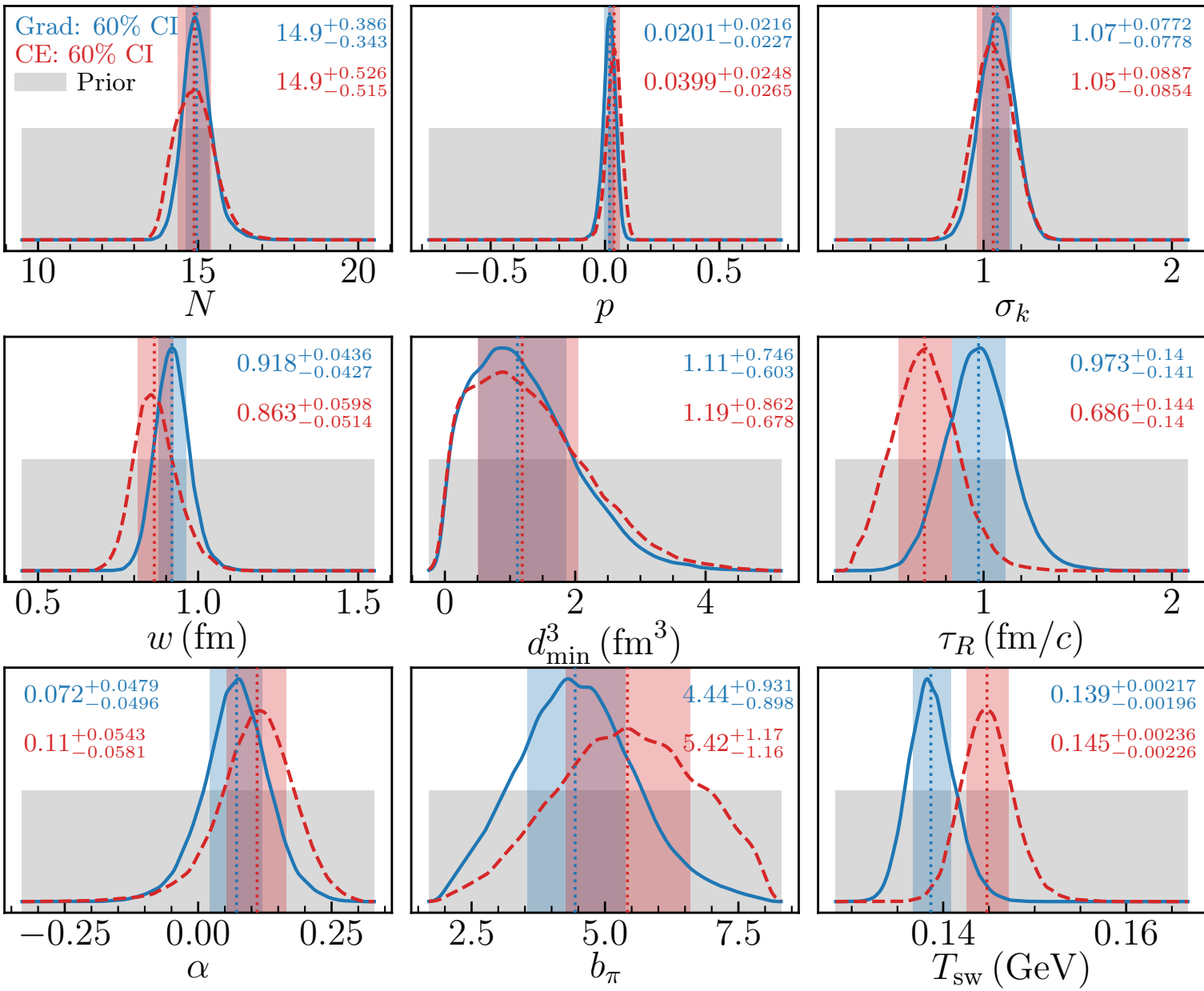
Bayesian parameter inference: Pb+Pb collision at 2.76 TeV

[SJ, PLB 874, 140243 (2026)]



$$y_{\text{exp}}(x) = \underbrace{\eta_{\text{mod}}(x, \theta)}_{\text{Predictions from approximate theory}} + \underbrace{\delta_{\text{MD}}^0(x)}_{\text{Error of approximate theory predictions}} + \underbrace{\epsilon(x)}_{\text{Observation error}}$$

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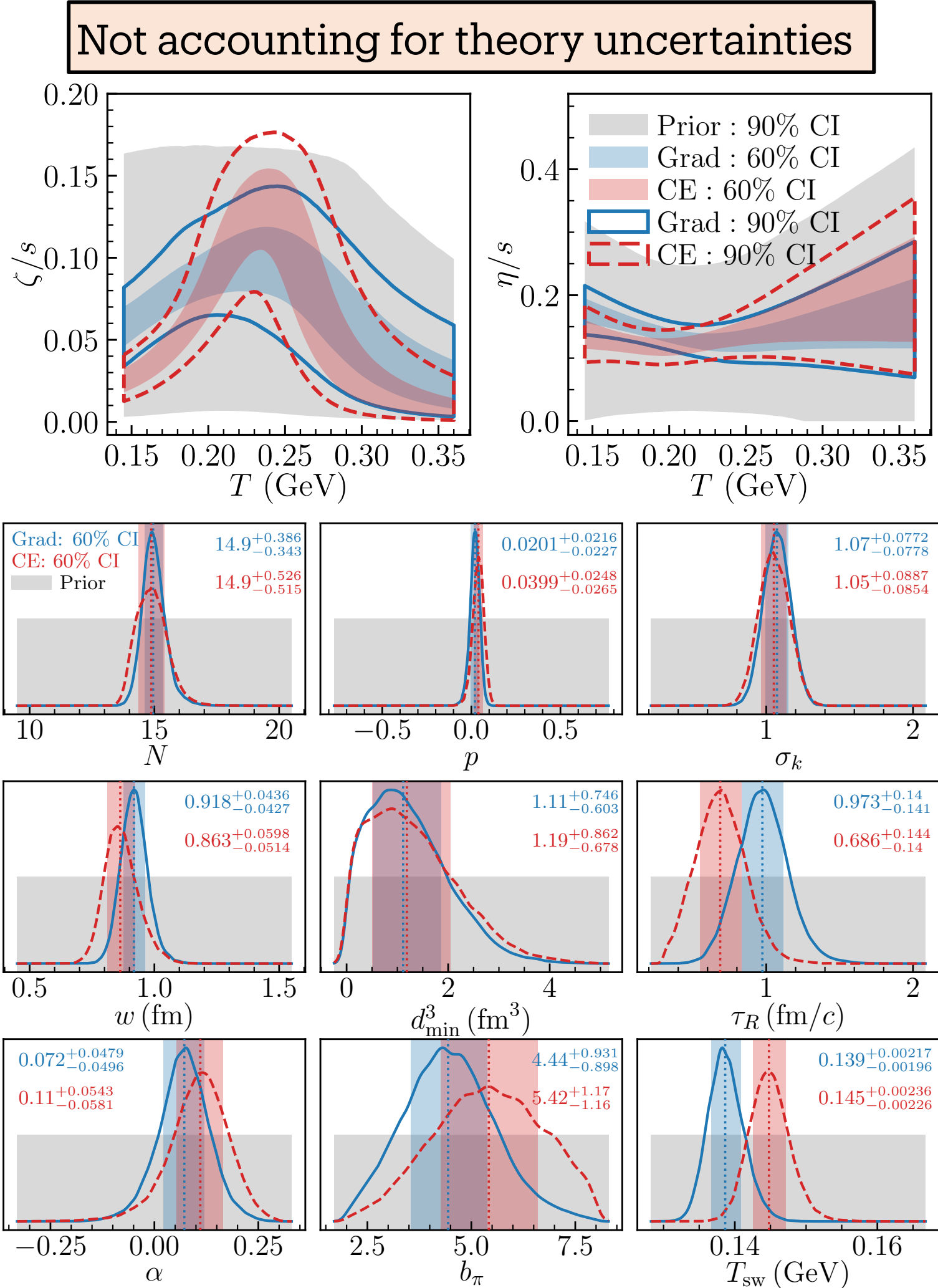
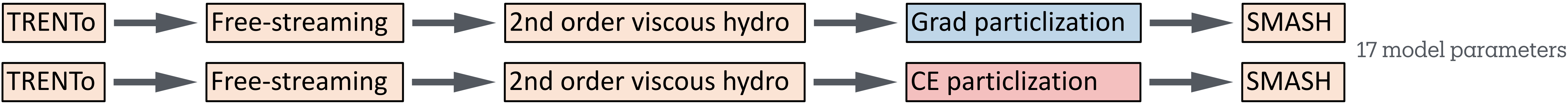
• Specific bulk viscosity posteriors differ between Grad (blue) and CE (red). Differences also visible for other parameters.

Bayesian updating of knowledge

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Bayesian parameter inference: Pb+Pb collision at 2.76 TeV

[SJ, PLB 874, 140243 (2026)]



Prior knowledge about theory domain (for both):
Multistage JETSCAPE model is more valid at small centralities (x) than at large centralities.
 $\Rightarrow$ **Theory is more correct at small x than at large x**
 $\Rightarrow \delta_{\text{MD}}(x)$ **should be non-decreasing with x**

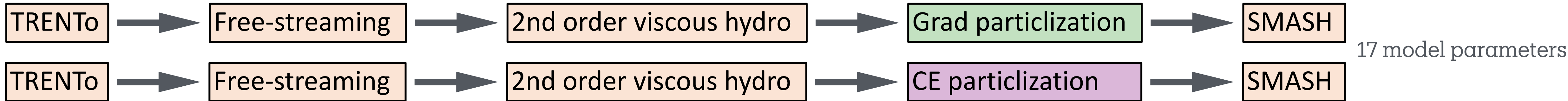
$$K(x_i, x_j | \phi) \equiv s^2 + \bar{c}^2(x_i x_j)^r \exp\left(-\frac{\|x_i - x_j\|^2}{2\ell^2}\right)$$

- Specific bulk viscosity posteriors differ between Grad (blue) and CE (red). Differences also visible for other parameters.

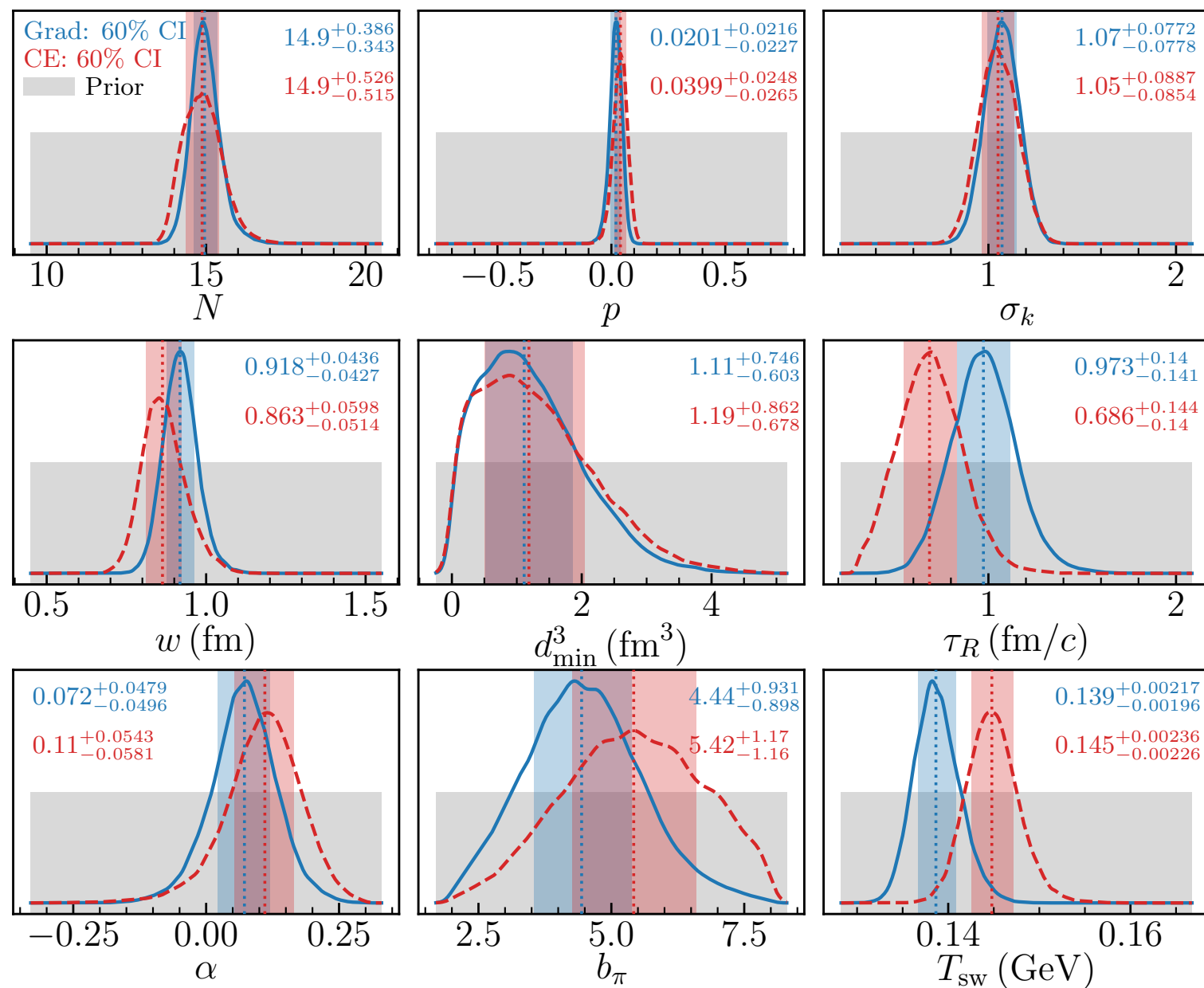
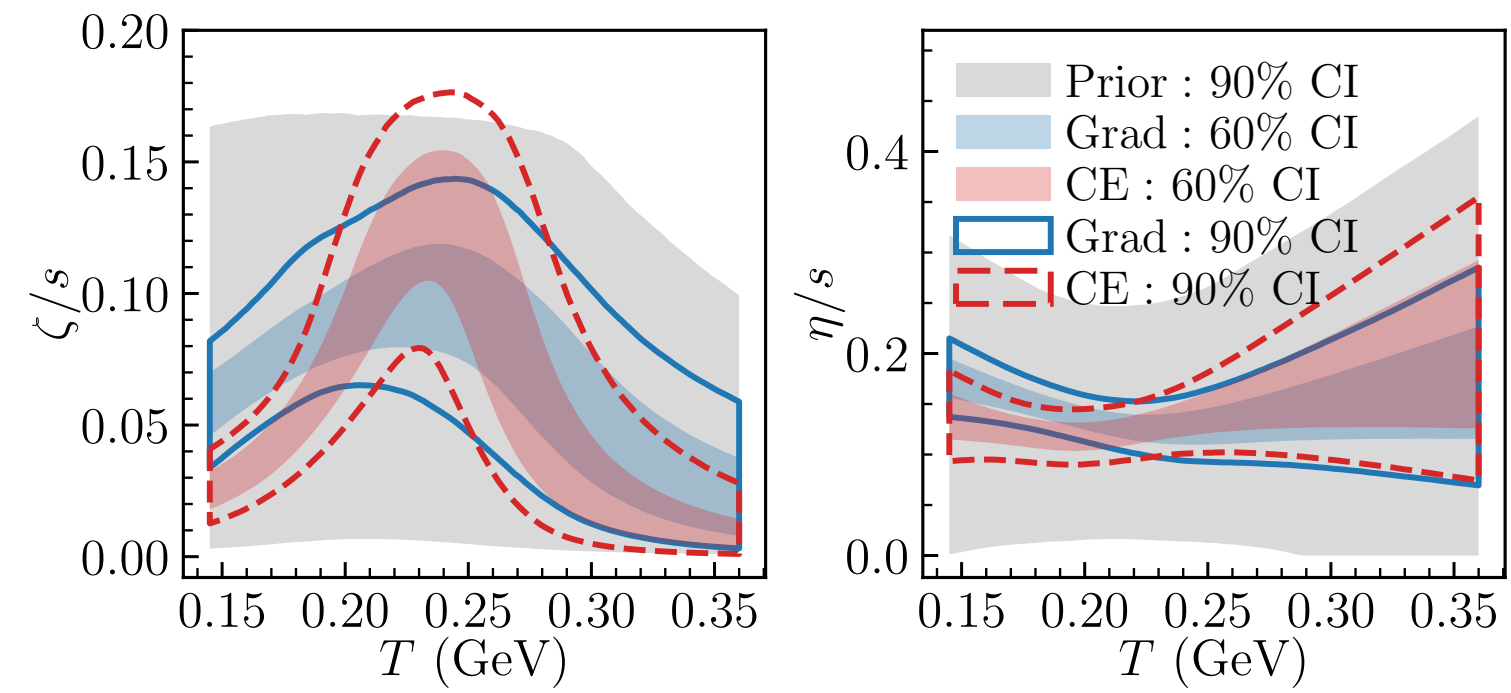
[D. Everett *et al.* 2010.03928, 2011.01430]

Bayesian parameter inference: Pb+Pb collision at 2.76 TeV

[SJ, PLB 874, 140243 (2026)]



Not accounting for theory uncertainties



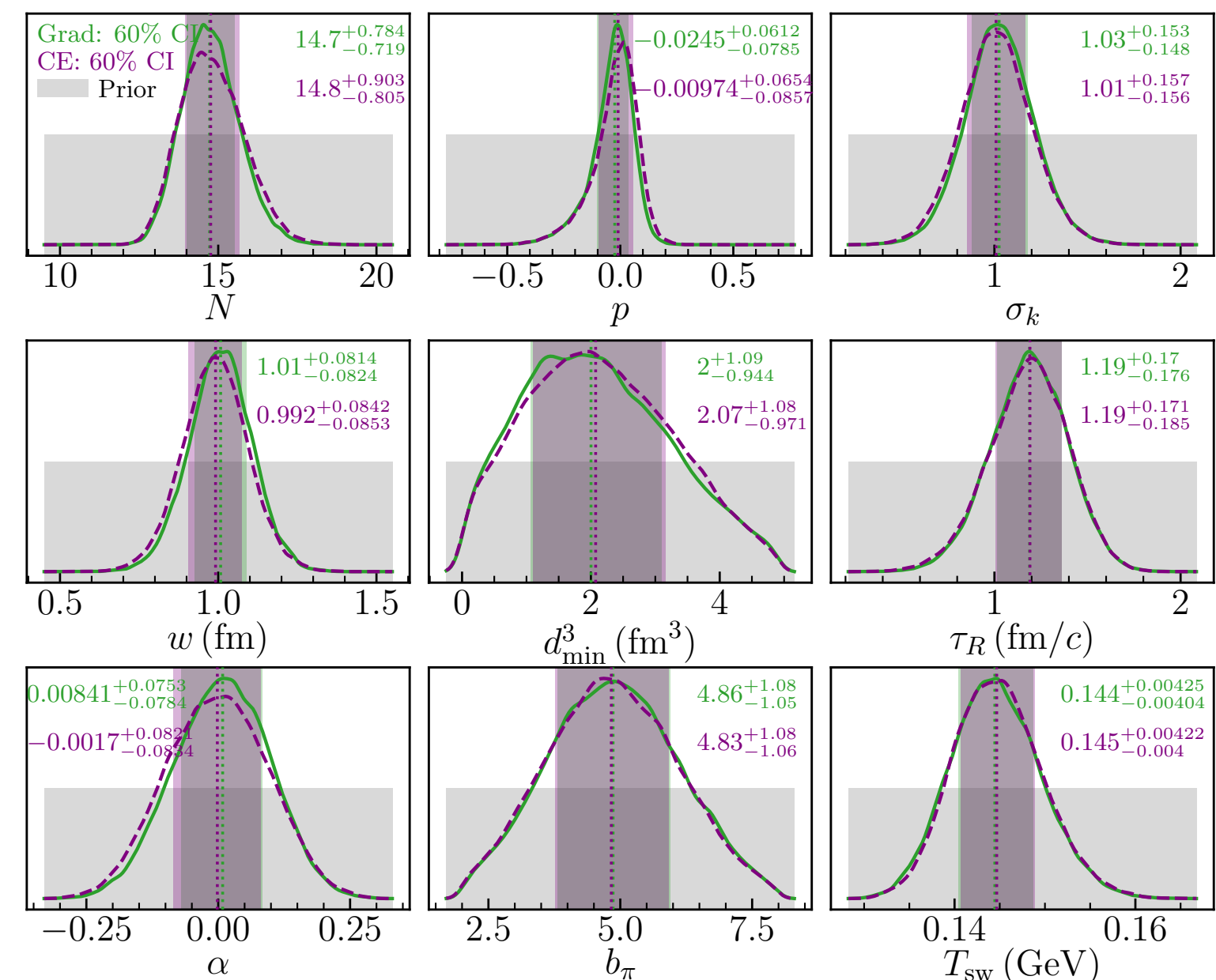
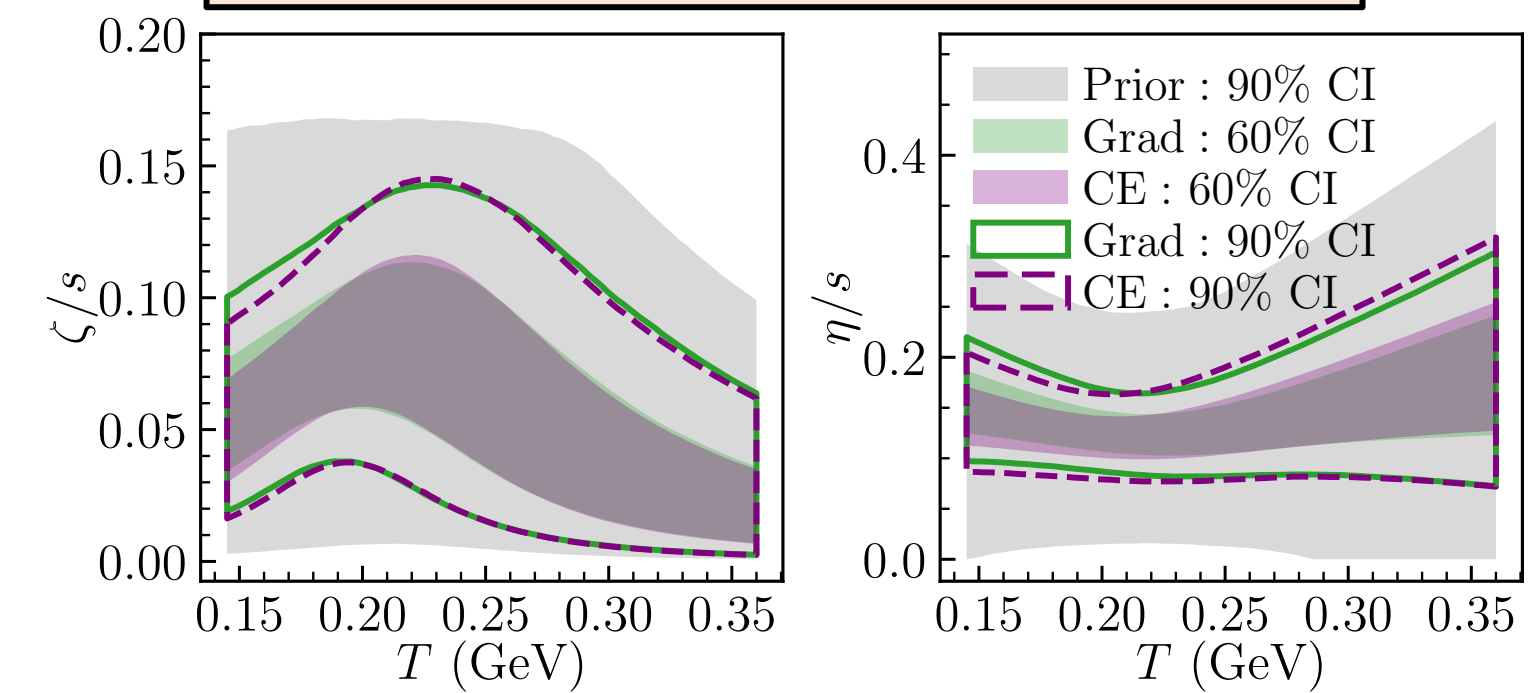
[D. Everett et al. 2010.03928, 2011.01430]

Prior knowledge about theory domain (for both):
Multistage JETSCAPE model is more valid at small centralities (x) than at large centralities.
=> Theory is more correct at small x than at large x
=> $\delta_{MD}(x)$ should be non-decreasing with x

$$K(x_i, x_j | \phi) \equiv s^2 + \bar{c}^2(x_i x_j)^r \exp\left(-\frac{\|x_i - x_j\|^2}{2\ell^2}\right)$$

- Specific bulk viscosity posteriors differ between Grad (blue) and CE (red). Differences also visible for other parameters.
- Posteriors are almost indistinguishable for all parameters for Grad (green) and CE (purple).
- Error in theory predictions due to approximate or missing physics is quantified and does not propagate across stages.

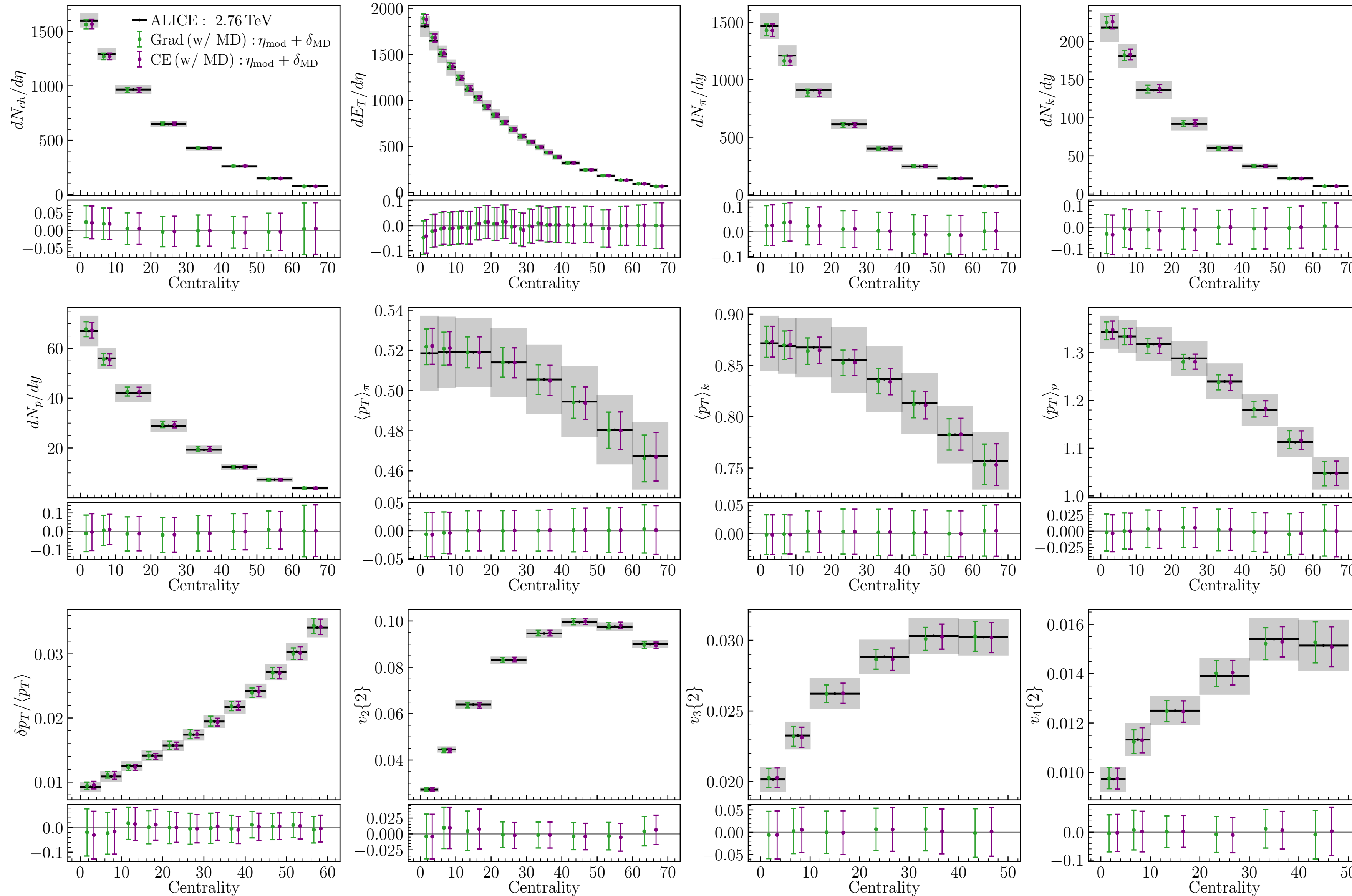
Accounting for theory uncertainties



[SJ, PLB 874, 140243 (2026)]

JETSCAPE model + discrepancy predictions

[SJ, PLB 874, 140243 (2026)]



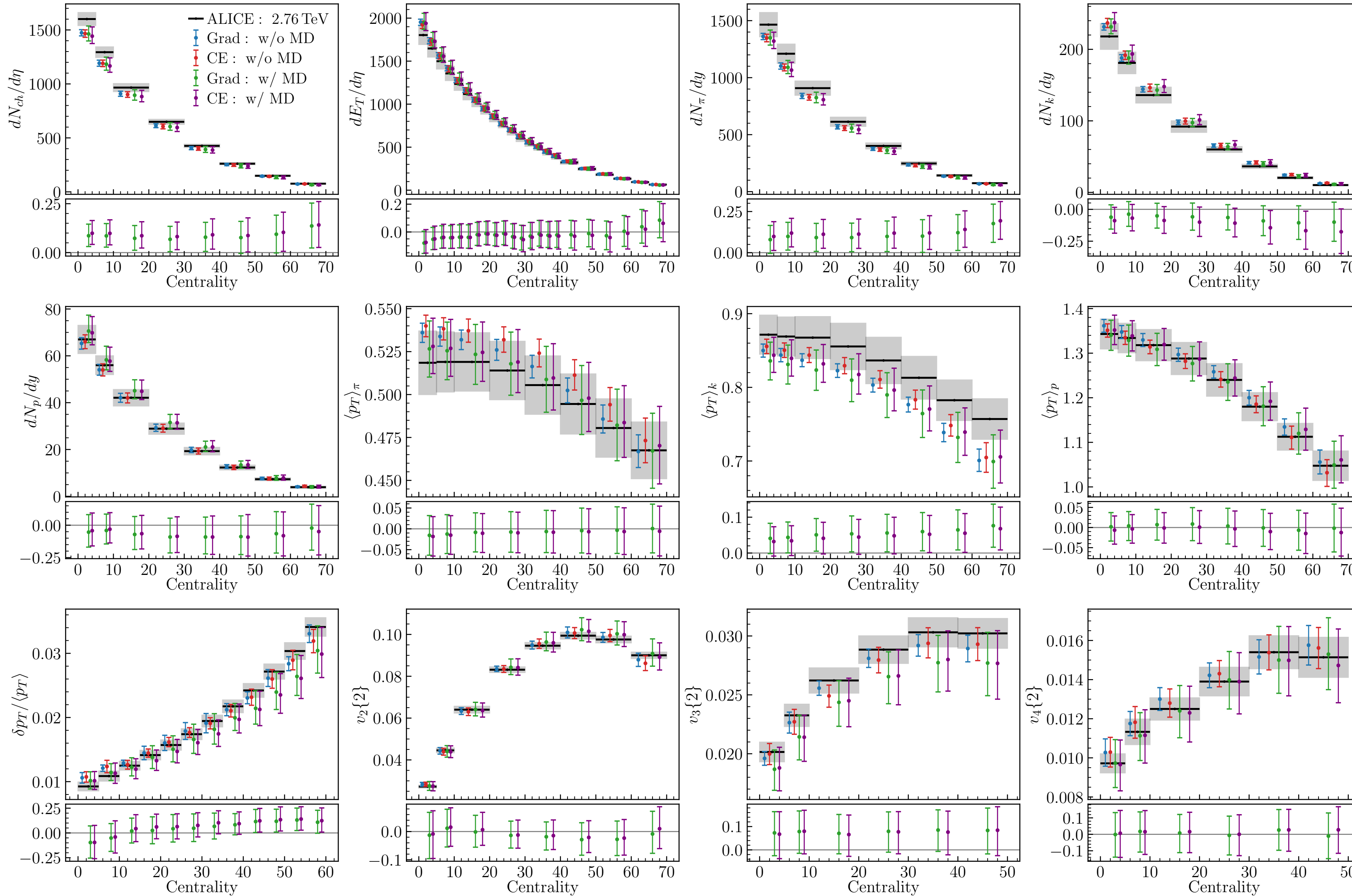
Predictions from
 $\xi(x; \theta, \phi) \equiv \eta_{\text{mod}}(x, \theta) + \delta_{\text{MD}}(x, \phi)$

Excellent agreement with data.
 Expected.

← normalized error $(y_{\text{exp}} - \eta_{\text{mod}} - \delta_{\text{MD}}) / |\langle y_{\text{exp}} \rangle|$

JETSCAPE model predictions

[SJ, PLB 874, 140243 (2026)]



$$y_{\text{exp}}(x) = \underbrace{\eta_{\text{mod}}(x, \theta)}_{\text{Predictions from approximate theory}} + \underbrace{\delta_{\text{MD}}(x)}_{\text{Error of approximate theory predictions}} + \underbrace{\epsilon(x)}_{\text{Observation error}}$$

- Both **Grad** and **CE** explains data equally good/bad.
- Theory predicts many observables correctly => learned $\delta_{\text{MD}}(x)$ are small for these.
- Deviations observed in w/ MD predictions => learned $\delta_{\text{MD}}(x)$.
- Large discrepancy in $\delta p_{\text{T}}/\langle p_{\text{T}} \rangle$ and $v_3\{2\}$ indicates deficiencies in initial condition model.**

← normalized discrepancies $(y_{\text{exp}} - \eta_{\text{mod}})/|y_{\text{exp}}| \approx \delta_{\text{MD}}(x, \phi)$

Comments about the DiscrepancyModel plots

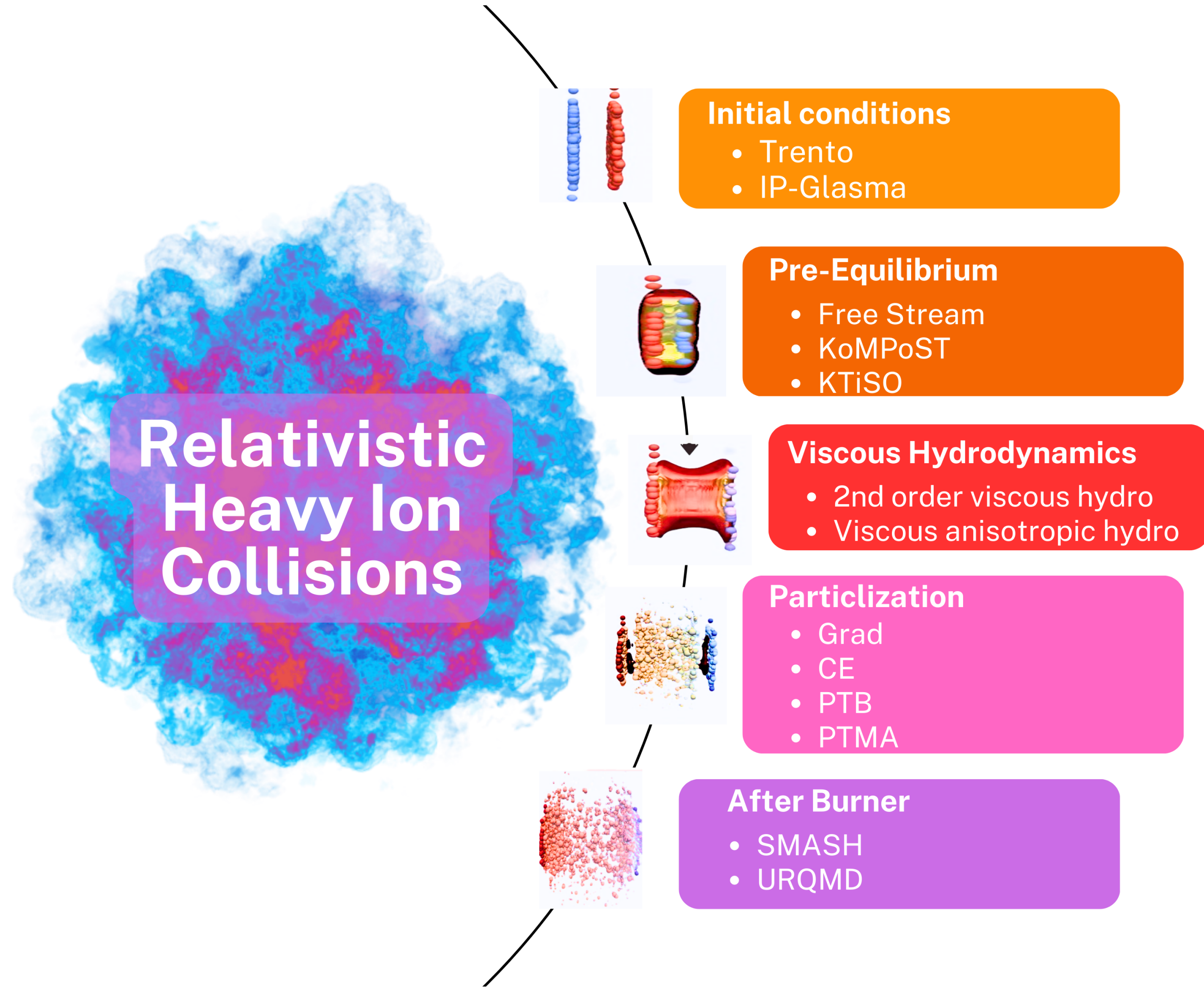
- blah, blah, blah

Summary

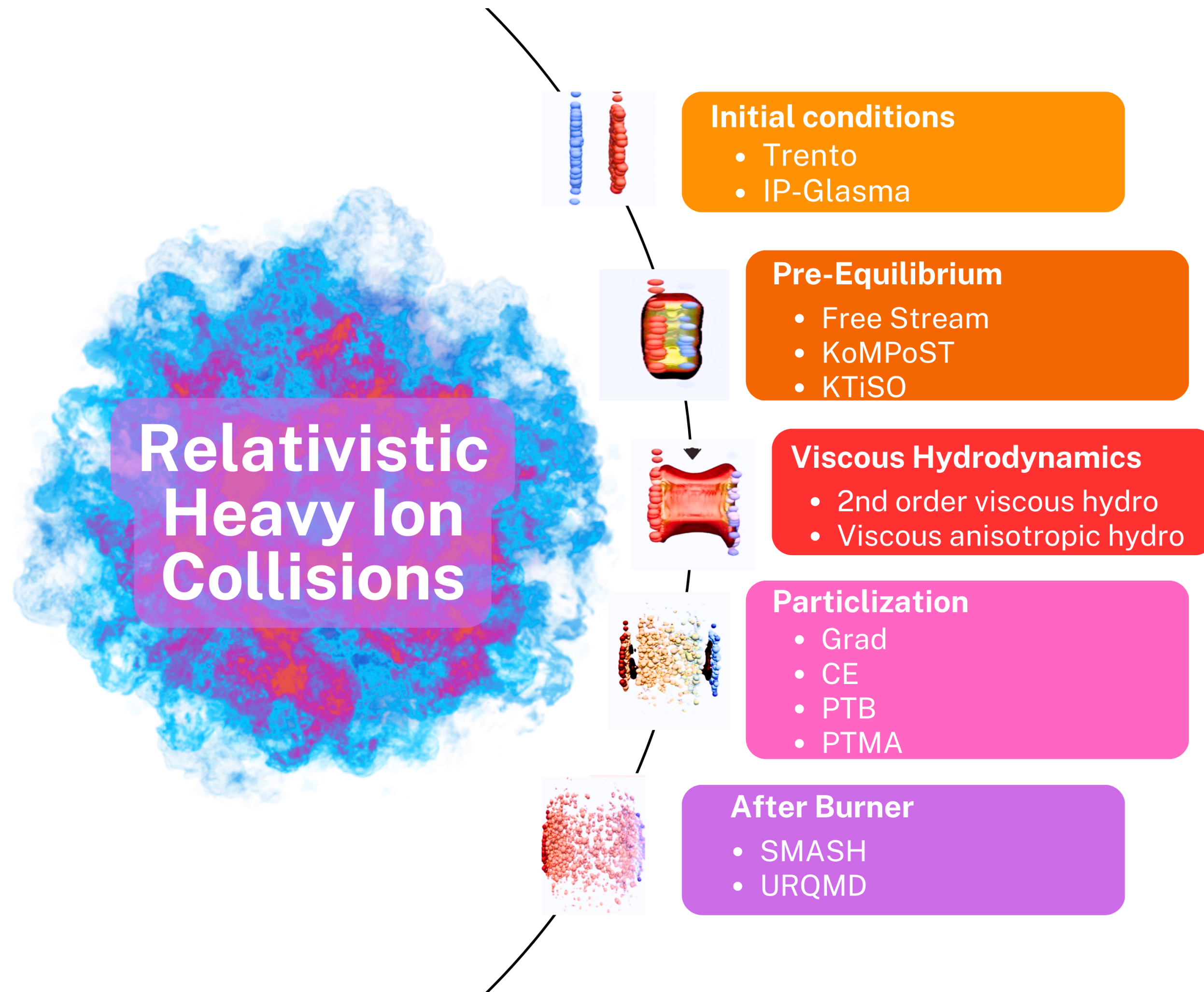
- Often the goal of model-data comparisons is to extract *physically meaningful* model parameters, **not to fit data**.
- **Quantifying theoretical uncertainties is crucial** for correct inference of *physical* model parameters.
- We demonstrated a **Bayesian framework that explicitly quantifies theoretical uncertainties** by statistically modeling theory uncertainties, guided by qualitative knowledge of the theory's domain of reliability (KOH/BOH).
- **We demonstrated functional prior construction using non-stationary GP for discrepancy function.**
- BAND Camp tutorial: <https://github.com/sjaiswal-tifr/ISNET12-model-discrepancy>
- Main repository of model discrepancy framework: <https://github.com/sjaiswal-tifr/ModelDiscrepancy>

Extra slides

Multi-stage physics model



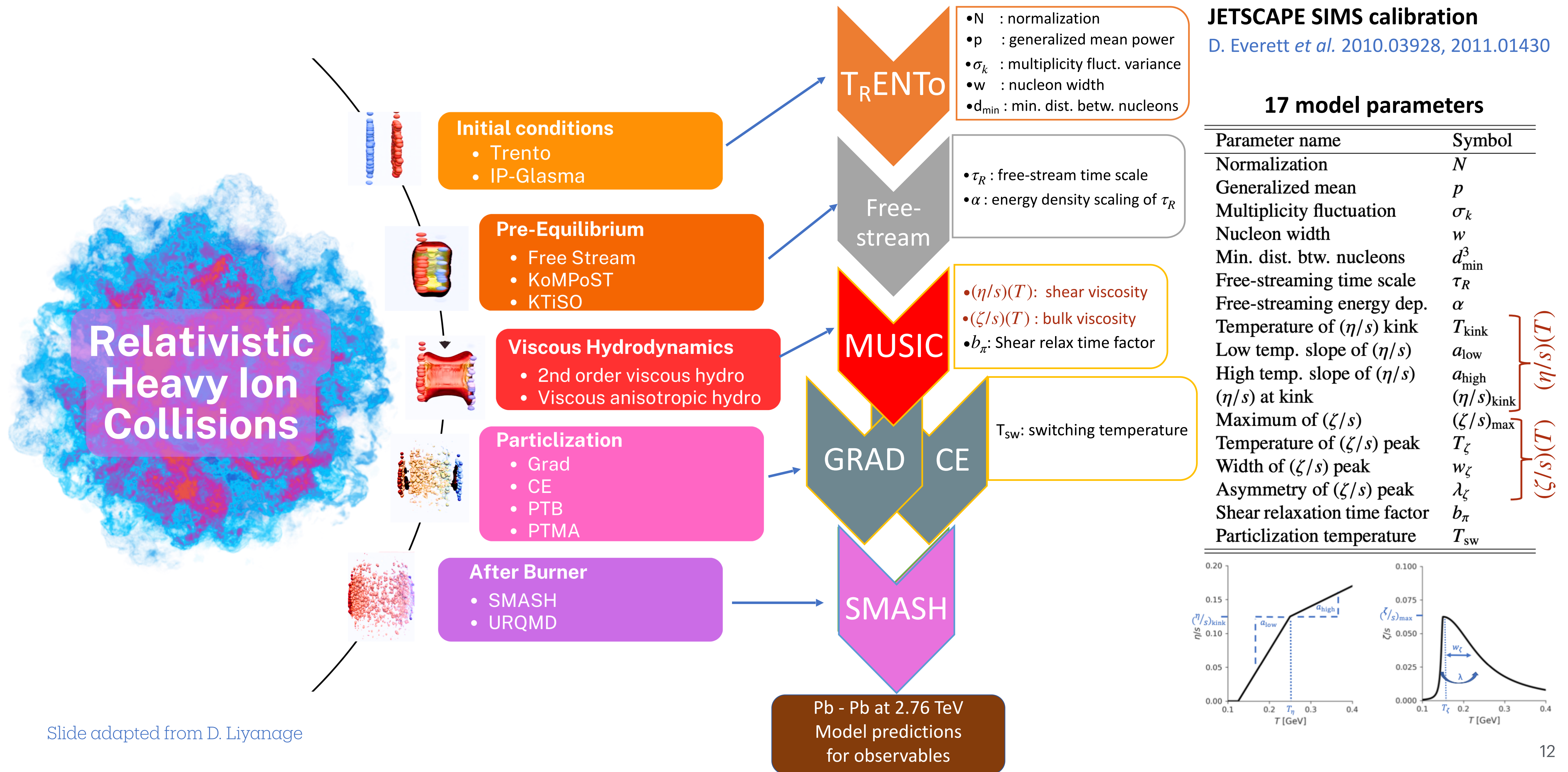
Multi-stage physics model



All models are approximate with associated theoretical errors.

Errors from one stage can propagate to other stages.

Multi-stage physics model



Modeling theoretical uncertainty (error) in heavy-ion

- Statistical model:
- Model $\delta(\cdot | \phi) \sim \text{GP}(\mathbf{0}, K(\cdot, \cdot | \phi))$

$$\underbrace{y(x_i)}_{\text{Experimental observation}} = \underbrace{\eta_{\text{mod}}(x_i, \theta)}_{\text{Theoretical prediction}} + \underbrace{\delta_{\text{MD}}(x_i, \phi)}_{\text{Theoretical error}} + \underbrace{\epsilon(x_i)}_{\text{Observation error}}$$

Prior knowledge about theory error: **JETSCAPE model is more reliable at small centralities than at large centralities.**

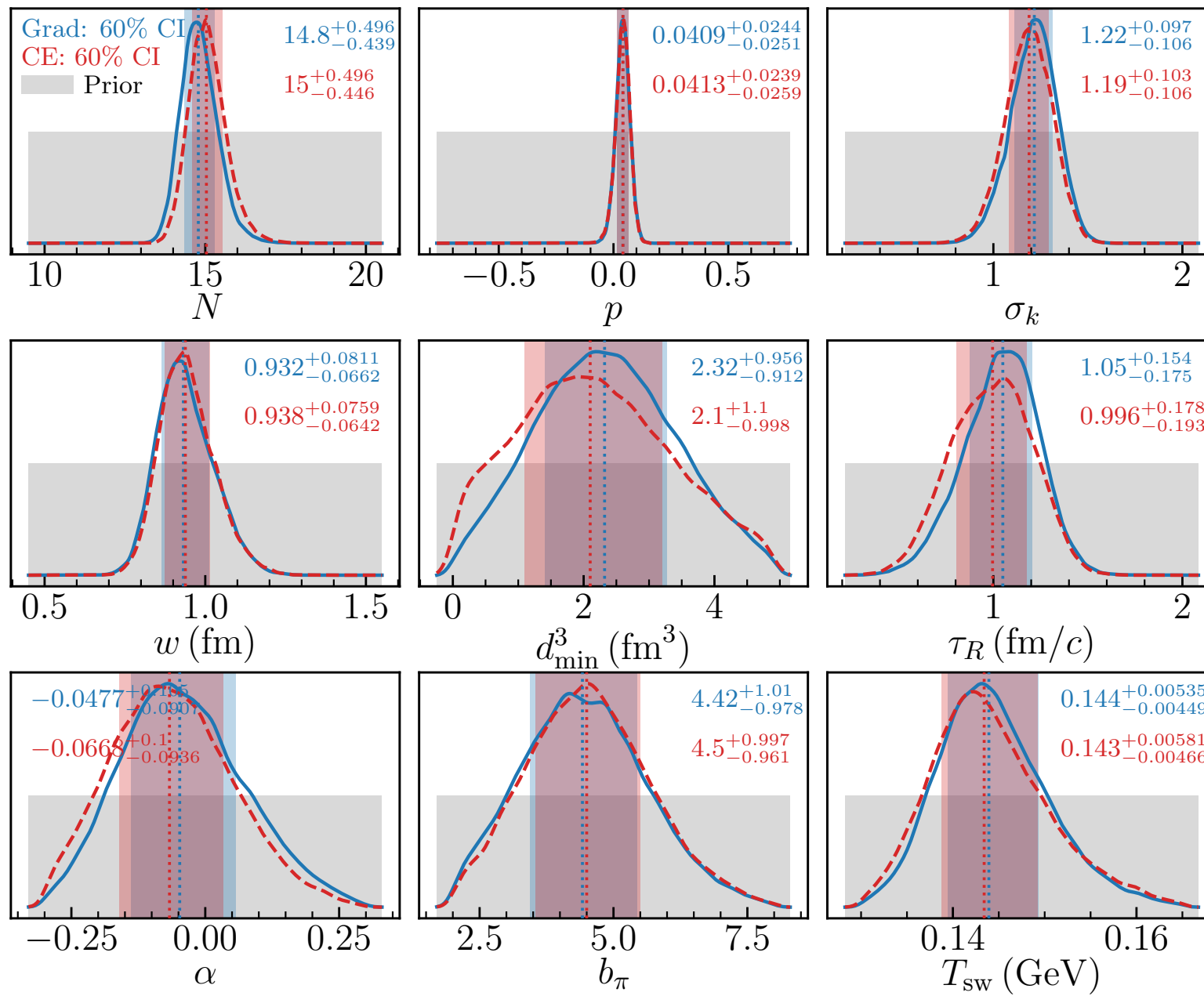
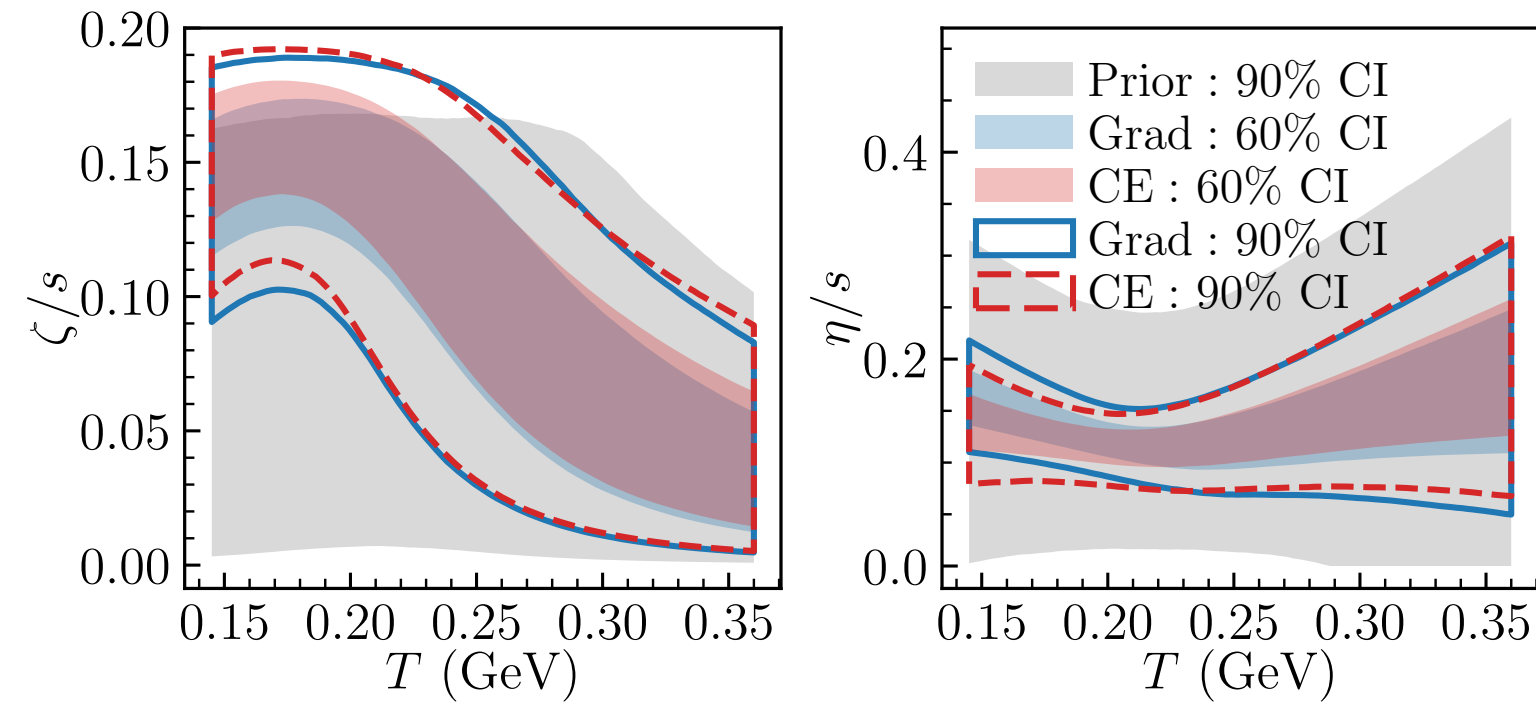
Reliability of hydrodynamics degrades as one moves from the most central to more peripheral collisions (smaller overlap). Also, Grad and Chapman–Enskog particlization schemes perform better in more central collisions (near-equilibrium corrections).

- Consider the kernel: $K(x_i, x_j | \phi) \equiv s^2 + \bar{c}^2(x_i x_j)^r \exp\left(-\frac{\|x_i - x_j\|^2}{2\ell^2}\right)$, ($r \geq 0$).
 - ➔ Allows theory errors to be constant or increase with centrality (but not decrease).
 - ➔ Non-vanishing theory error allowed even for ultra-central collisions ($s \neq 0$).
 - ➔ $s = \bar{c} = 0 \implies \delta_{\text{MD}} = 0$.

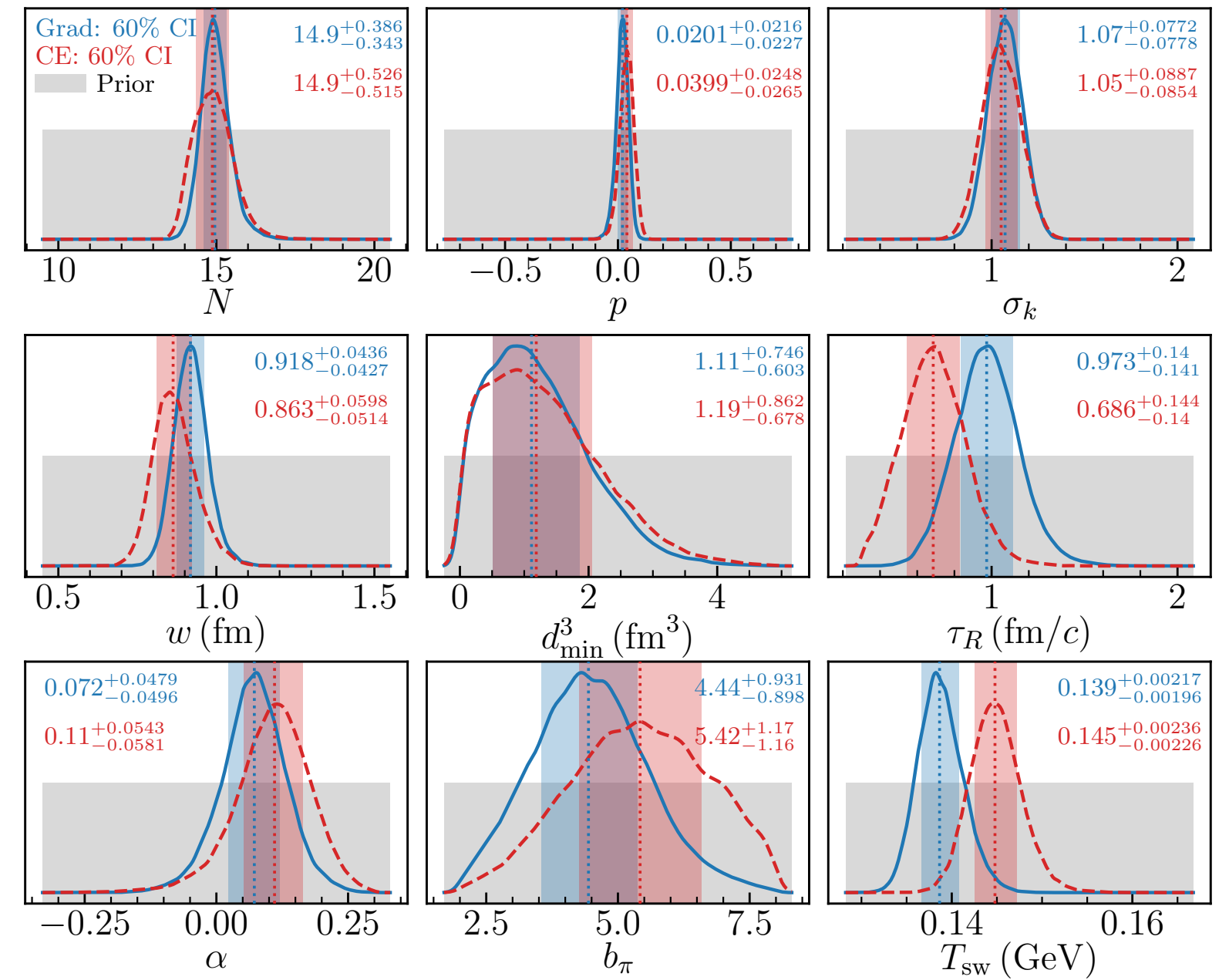
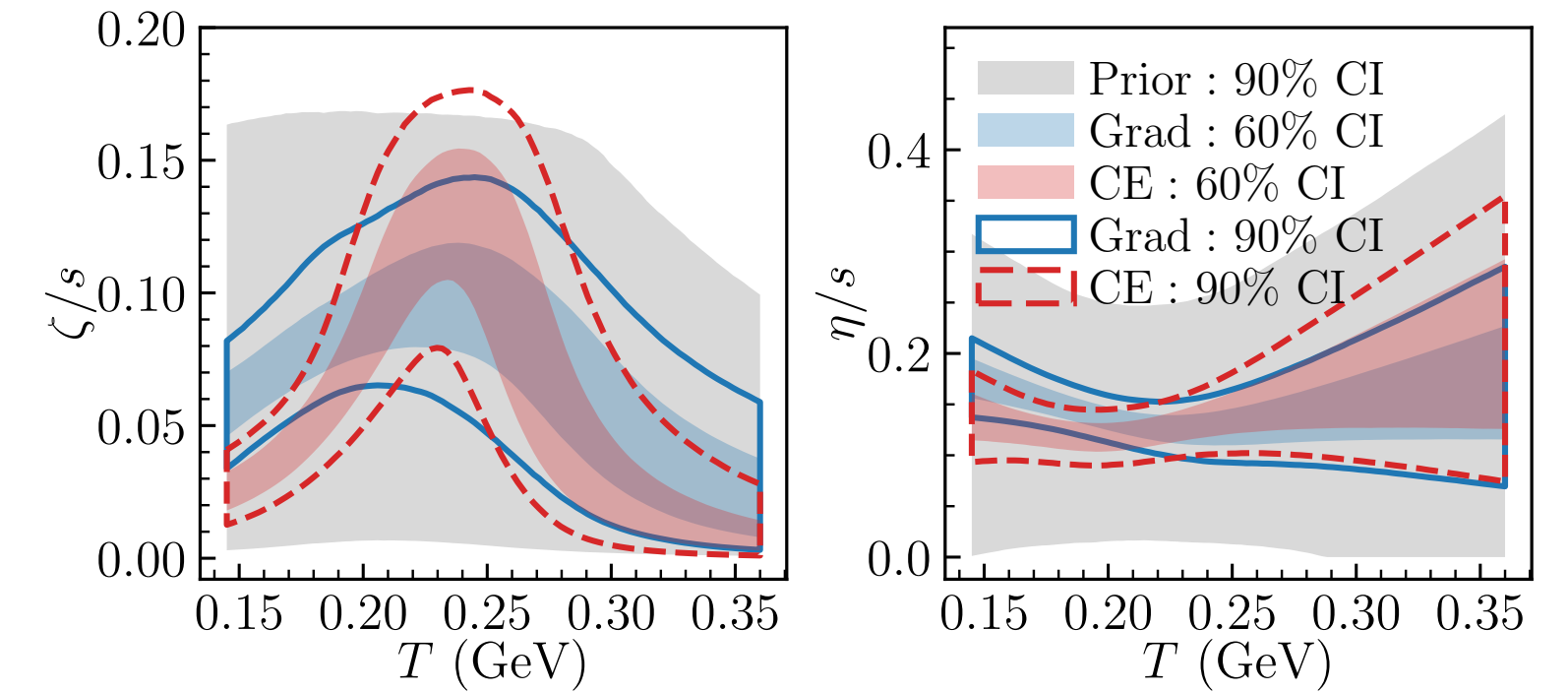
Check (w/o MD): Stability of posteriors

SJ, arXiv: 2509.19759

Calibration with subset of observables:
 $dN_{ch}/d\eta$, $dE_T/d\eta$, and $v_n\{2\}$ ($n = 2,3,4$)



Calibration with all observables



w/o MD

Posteriors for many parameters shift significantly when all measurements are included in the inference, indicating lack of robustness of the parameter estimate.

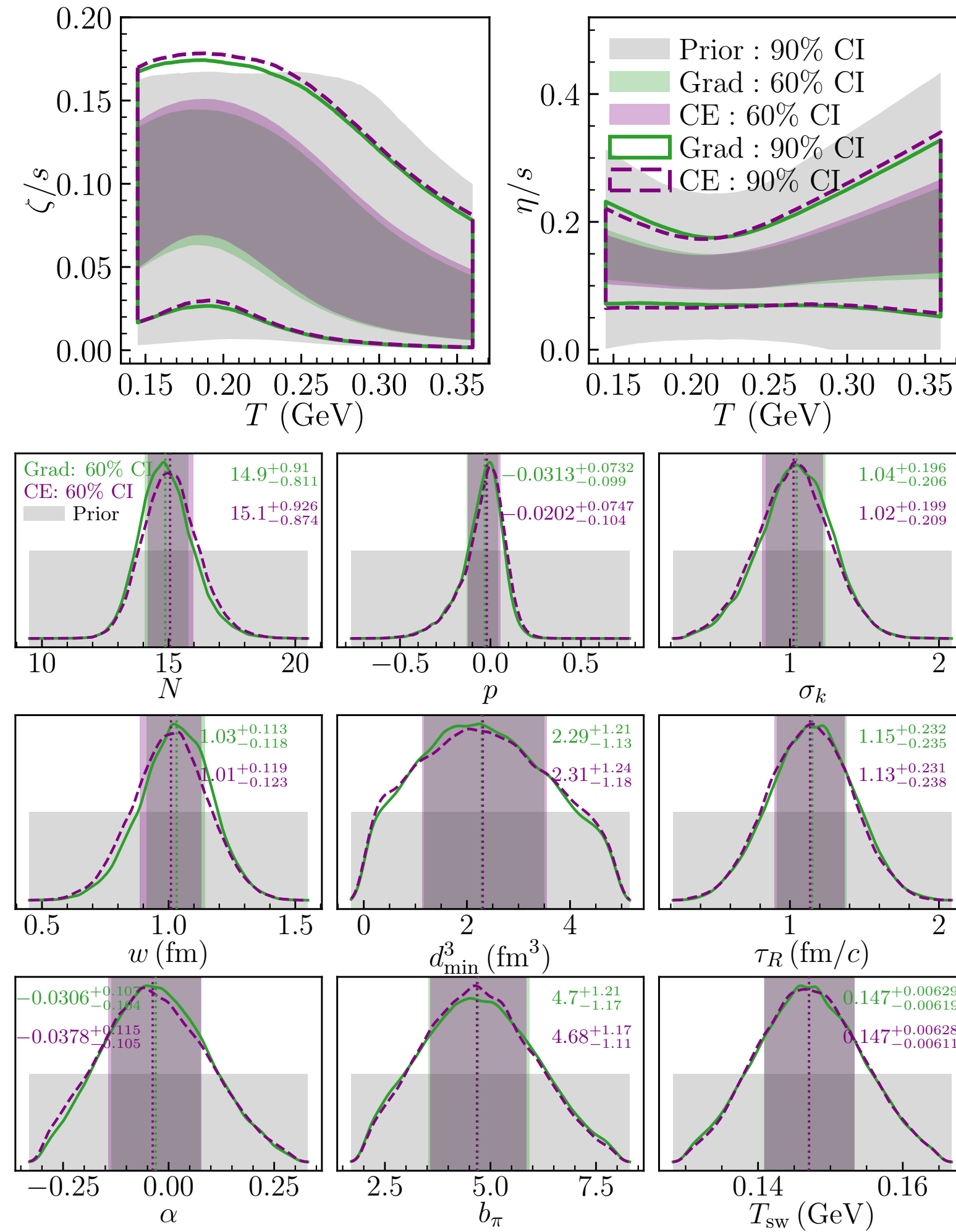
Check (w/ MD): stability of posteriors

SJ, arXiv: 2509.19759

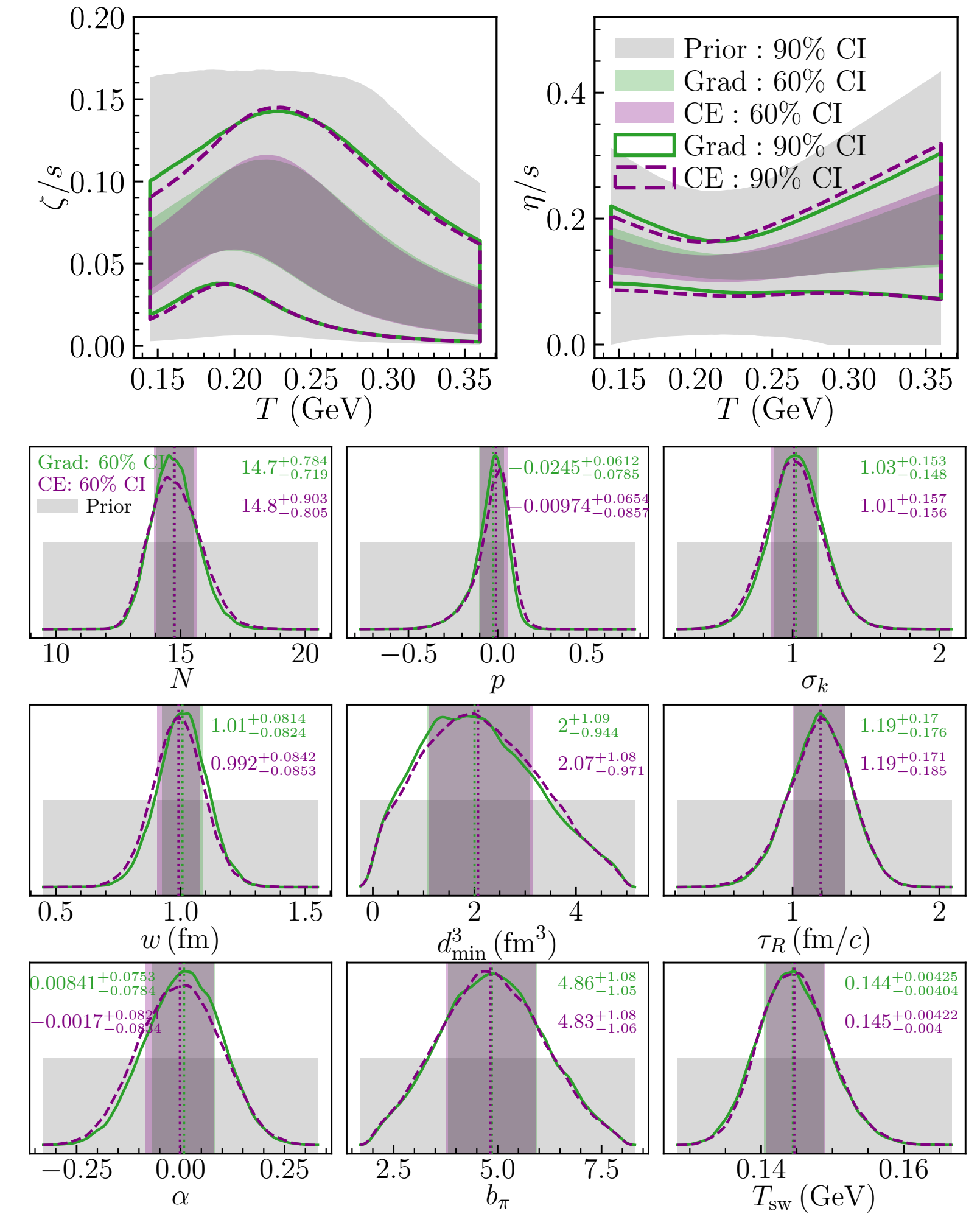
Calibration with subset of observables:
 $dN_{ch}/d\eta$, $dE_T/d\eta$, and $v_n\{2\}$ ($n = 2,3,4$)



Calibration with all observables



w/ MD



Posteriors tighten without shifting, indicating robust parameter inference.